

Understanding Disparities in Punishment: Regulator Preferences and Expertise

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This paper quantifies the benefits of discretion in the enforcement of environmental regulations. We identify and estimate a structural model of regulator-discharger interactions, exploiting an increase in the enforcement stringency of water pollution regulations in California. Our estimates indicate that most of the heterogeneity in punishments for observably similar violations is due to heterogeneity in discharger compliance costs rather than heterogeneity in regulator preferences. We find that removing the discretion of regulators to tailor punishments to discharger attributes would raise enforcement costs and decrease compliance by dischargers with high social harms of violations.

I. Introduction

Regulations are often written flexibly so that the authorities in charge of enforcing them can do so judiciously. For example, following a particular violation, enforcement authorities may be able to choose from a range of

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possible punishments, considering a host of aggravating and mitigating factors that are at times subjective. One reason to allow discretion in enforcement is that it might be impossible, in practice, for the written regulation to specify all possible contingencies—especially when the circumstances surrounding any given violation, such as the compliance costs borne by the regulated entity and the resource constraints faced by the enforcement authority, can vary considerably from one case to another. However, without proper incentives, regulators may pursue their own interests rather than apply their expertise to make an appropriate judgment as a social planner (Stigler 1971; Peltzman 1976). In this paper, we identify and estimate a game-theoretic model of regulation to empirically evaluate regulatory discretion in the context of the enforcement of water quality regulation in California.

To begin with, we argue that the enforcement policies of the Clean Water Act and California's Porter-Cologne Water Quality Control Act leave room for regulatory discretion, giving rise to concerns over inconsistencies in the enforcement of these laws within the state. We then provide empirical evidence that regulators use their discretion, by documenting disparities in penalties—that is, variation in penalties, conditional on violation attributes. To do so, we employ our unique data containing the universe of recorded water quality violations in 2000–2014, as well as all monetary penalties associated with each individual violation. By linking violations and penalties, these data allow us to gauge regulation enforcement standards, without having to make an assumption on the time lag between violations and enforcement actions.¹ In our analysis, we focus on domestic wastewater treatment facilities, which are among the most important violators of water quality regulation in California. We find that the penalty disparities are correlated with various facility and local attributes, such as the facility's size and age, and demographic, economic, and environmental attributes of the location. In addition, the expected penalty for a violation increases when there are other concurrent violations by the same facility.

Becker (1968) provides a theoretical justification for disparities in punishment, by considering the optimal amount of enforcement when it is costly to impose sanctions. Under this framework, disparities in punishment may have a few nonexclusive sources: heterogeneity in the private

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¹ This feature of our data is an important advantage, relative to the data sets employed by the existing literature on the enforcement of the Clean Water Act (Magat and Viscusi 1990; Earnhart 2004a, 2004b; Shimshack and Ward 2005; Gray and Shimshack 2011), where the link between an enforcement action and a violation is not observed.

violation gains, the social harms of violations, and the costs of punishment. Our structural approach helps with disentangling these sources of disparities and better understanding regulatory discretion. We adopt the model of Mookherjee and Png (1994), which generalizes Becker (1968) and incorporates the main institutional features of our setting: facilities choose a continuous (as opposed to binary) level of effort to reduce violations; the regulator can impose punishment at varying rates, depending on the level of offense; and facilities may have private information on the costs of complying with regulation.

In the model, for each facility, the regulator sets a penalty schedule. Given these schedules, the facilities exert costly effort to affect the extent of compliance. The regulator knows the distribution of compliance cost types of each facility but observes neither the realized types nor the facilities' efforts, so the penalty is a function of the violations only. In determining the penalty schedules, the regulator minimizes the sum of the facilities' expected compliance costs, the cost from environmental hazards of violations, and the enforcement costs associated with imposing penalties. Departing from the normative framework of Becker (1968) and Mookherjee and Png (1994), we allow the latter two costs to partially reflect the regulator's private concerns, and we refer to these costs as the regulator preferences.

We use data on violations and penalties to identify and estimate the model, exploiting a set of institutional changes in the mid-2000s that were aimed at reducing the administrative burden of enforcement and making the compliance information more accessible to the public. These changes include the launch of a new computerized system to track and manage information about violations and enforcement and the establishment of a new statewide office to support enforcement activities. We document that, after these changes, the average amount of penalties increased and violations became less frequent. Moreover, we find little evidence that the compliance cost structure changed during the period. This institutional feature provides a unique opportunity to identify the dischargers' compliance cost function.

We provide conditions under which the model is nonparametrically identified. Our identification strategy is closely related to two recent papers, d'Haultfoeuille and Février (2020) and Luo, Perrigne, and Vuong (2018), both of which address the identification and estimation of adverse selection models. The former study focuses on the informed party and employs exogenous variation in the contracts (which, in our application, is provided by the mid-2000s institutional changes) to identify that party's distribution of types (the distribution of facilities' compliance costs). This approach does not rely on the regulator's optimality. Conversely, the latter paper builds on the optimality conditions of both the informed and the noninformed parties to identify the model primitives without

necessarily relying on any external variation. By combining both approaches, we identify a more general model than the ones considered by these two papers. We develop a multistep estimator that closely follows the constructive identification strategy. In this regard, our paper relates to the structural empirical literature on regulation (Wolak 1994; Thomas 1995; Gagnepain and Ivaldi 2002; Timmins 2002; Brocas, Chan, and Perrigne 2006; Ryan 2012; Gagnepain, Ivaldi, and Martimort 2013; Oliva 2015; Fowlie, Reguant, and Ryan 2016; Lim and Yurukoglu 2018; Abito 2020).

We find suggestive evidence that the regulators tailor the enforcement policies to local residents' preferences. Specifically, the estimation results indicate that the regulators tend to consider the violations by a facility located in a county with a high average household income as more environmentally damaging and less costly to punish than those by other facilities. To the extent that environmental goods or amenities are a normal good, this finding is consistent with the idea that regulators may reflect local preferences on water quality when they enforce the water regulation. Moreover, we find that the estimated enforcement costs are negatively correlated with the percentage of voters supporting California Proposition 84 in 2006, which authorized the issuing of bonds to fund water quality improvement projects at the county where the facility is located.

Given our estimates of both the compliance cost structure and the regulator preferences, we quantify the extent to which the heterogeneity in regulator preferences explains the observed disparities in penalty stringency. In doing so, we consider a hypothetical scenario in which all facilities are subject to a regulator with the same preferences—set at the median across our preference estimates. Not surprisingly, we find that, relative to the current regime, homogenizing the regulator's preferences would reduce the dispersion of penalty stringency across facilities. However, this reduction is by a moderate extent—11%.

We also assess the implications of limiting regulatory discretion. As discussed earlier, our descriptive data analyses reveal two main channels through which regulatory discretion in enforcement manifests itself: penalty adjustments that vary with (i) facility and local attributes and (ii) the violator's conduct in terms of other violations during the period. In light of these findings, we consider two counterfactual scenarios, limiting each of these channels at a time. First, we evaluate a one-size-fits-all policy, where all facilities face the same penalty schedule. Then we examine a linear policy, in which the regulator is constrained to set a per-violation penalty amount. Here we assume that the regulator chooses penalty schedules that minimize the sum of the costs from compliance, environmental hazards, and enforcement across all facilities, under the constraint imposed by each scenario. We find that violations on average would slightly decrease under both scenarios (roughly 6%) but at the

expense of an increase in total penalties. More importantly, we measure the penalty savings associated with allowing for each type of regulatory discretion; for example, we show that a linear penalty policy that would induce facilities to adhere to the compliance level of the full discretion scenario would lead to 10% higher penalties in total. We also find that implementing the one-size-fits-all policy would lead to more violations by facilities perceived to have high benefits of compliance, such as those posing a high threat to water quality.

One of the perils of regulatory discretion stems from the potential misalignment between the preferences of the regulator and those of the social planner. In the absence of external estimates of the latter, we cannot quantify the effects of such misalignment on enforcement and compliance. Instead, we assess an upper bound of the extent of violations that might be associated with inappropriately low perceptions of the benefits of compliance by the regulator, possibly driven by corruption or lack of dedication. To do so, we consider an alternative regulator who highly values the environmental benefits of compliance and perceives enforcement costs as minimal for all facilities. We find that under such a scenario, the number of violations would fall by half and the assigned penalties would increase by 77%.

Our paper contributes to the empirical literature on regulatory mechanisms and the incentives of regulators. Existing studies show evidence on various determinants of enforcement stringency, including local economic conditions, public health risk, pressure from special interest groups, the political ideology of the government, and the agency budget.² Because the benefits and costs of a regulation are not directly observed, it is difficult to empirically evaluate the regulatory mechanism using a reduced-form approach. Our analysis, along with recent papers by Duflo et al. (2018) and Blundell, Gowrisankaran, and Langer (2020), provides a step forward to this end, by identifying and estimating a regulation model. The former study uses data from a field experiment in India that doubled the inspection frequency for a random group of manufacturing plants and quantifies the extent to which regulators' discretion over what plants to target helps enforcement. The latter assesses the gains from the Clean Air Act's dynamic enforcement, where repeat offenders are placed in high-priority violator status and receive more frequent inspections and higher penalties. The results of these papers, consistent with ours, indicate that limiting regulatory discretion may increase penalties and enforcement costs. Unlike these two recent papers, we estimate regulator

² See Scholz (1986); Deily and Gray (1991); Cropper et al. (1992); Gray and Deily (1996); Helland (1998); Agarwal et al. (2014); Gordon and Hafer (2014); Holland (2016). Recently, Burgess, Olken, and Sieber (2012) and Jia and Nie (2017) provide suggestive evidence of regulatory capture in developing countries, and Leaver (2009) shows that regulators' desire to avoid public criticism can lead to inefficiency even without capture.

preferences. Having estimates of these preferences allows us to examine their determinants, investigate their role in explaining penalty disparities, and predict how regulators would respond to policies limiting their discretion.

The rest of the paper is organized as follows: section II describes how the water quality regulations in California are enforced and provides details of the institutional changes. In section III, we present the data and some descriptive statistics. Section IV contains the theoretical model, and section V describes the identification and estimation of the structural model. Section VI presents the estimation and counterfactual results. We conclude in section VII.

II. Institutional Background

A. Water Quality Regulation and Enforcement

Both the Clean Water Act and the state's Porter-Cologne Water Quality Control Act govern the water quality regulation in California. The former act created the National Pollutant Discharge Elimination System (NPDES) to regulate facilities that discharge pollutants from any point source, such as a pipe or a ditch, into surface waters, including lakes, rivers, and the ocean. Although the program is federal, the state government has administered it since the authorization by the Environmental Protection Agency (EPA) in 1973. An NPDES permit is typically a license for a facility to discharge a specified amount of a pollutant into a receiving water under certain conditions, where the limits on the concentration of the pollutants are based on both the availability of pollution control technologies and the water quality standards of the receiving water.

Both laws require that permittees periodically submit discharge monitoring reports with information about the quantity and quality of their effluents. They are required to sample receiving waters, perform bioassays, and measure and report the toxicity potential of the discharges. Enforcement actions are mostly based on these reports. Our data regarding all NPDES violations in the state during the period of 2000–2014 indicate that 95% of the recorded violations were detected from permittees' self-reports, while the remaining 5% were detected during an inspection or triggered by a complaint, referral, or sewer overflow.

The California Water Boards, consisting of the State Water Resources Control Board and nine Regional Water Quality Control Boards, are in charge of enforcing the water quality regulations in the state. The state board oversees the regional boards, which have primary jurisdiction in issuing permits, monitoring water quality, and taking enforcement actions against violating dischargers. Each regional water board consists of seven board members, who are appointed by the governor to serve 4-year terms

once they are confirmed by the state senate. The board members do not serve multiple regional boards simultaneously; they must reside in or have a principal place of business within the region that a given board covers. They serve part-time and conduct their business at regular meetings, which are normally held 10 times per year to make decisions on water quality matters.³ The board members rely on the staff—most of whom are engineers, geologists, and biologists—to conduct the day-to-day tasks associated with water quality management, such as setting water quality standards, drafting permits, and conducting enforcement activities.

B. Penalty Determination

For an initial determination of compliance, the regional board staff screens the self-monitoring reports. When a violation is identified, the staff issues a formal notice to clarify errors, vague permit language, or other areas of disagreement between the discharger and the staff. If a violation is confirmed, the case is then evaluated for enhanced enforcement, such as an administrative civil liability (ACL), which might result in a monetary penalty. To impose an ACL, the staff must make an ACL complaint, followed by a 30-day public comment period. The notice for the comment period is posted on the water board's website and may also be mailed to interested parties or published in a local newspaper. The discharger may waive its right to a board hearing and pay the liability, negotiate a settlement, or appear at the hearing to dispute the ACL.

The enforcement policy (California State Water Resources Control Board 2002, 2010) provides a detailed penalty calculation methodology. It involves determining the initial penalty amount and then adjusting it to address various factors. The initial penalty amount is based on the extent and severity of the violation, the sensitivity of the receiving water, and any impacts of the violation on beneficial uses of the affected water. The initial penalty amount is then adjusted to reflect a number of factors. Some of the factors are associated with the violator's conduct, such as whether the violation was accidental as opposed to intentional or negligent and whether there is a pattern of (intentional) repeat violations. The remaining adjustment factors are the violator's ability to pay and continue in business; the economic benefit of the violations; costs of investigation and enforcement, including any expert witness expenses; environmental justice issues related to whether the violations have a disproportionate impact on a particular disadvantaged group of people; and the maximum

³ Based on the description of each regional board member available online, we find that, on average, a regional board consists of four members from the private sector (e.g., a law firm partner or a farm owner), one or two members from the public sector (e.g., a mayor), and one or two members from academia.

and minimum liability amounts. After these adjustments by the water board staff, the final penalty amount may be further modified by the water board, possibly as a result of negotiations with the violator.

An important regulation associated with a minimum liability amount is that serious or multiple nonserious NPDES violations are subject to a mandatory minimum penalty (MMP) of \$3,000 per violation. A serious violation is associated with a discharge above limits of a group I (group II) pollutant by 40% (20%) or more.⁴ As for nonserious violations, the minimum penalty applies when such violations occur four or more times in any period of six consecutive months.

C. Regulatory Discretion

The enforcement policy provides guidelines for calculating the initial penalty amount and adjustments, but these guidelines leave room for discretion by the water board staff in defining and quantifying each factor in the calculation of the penalty amount. In addition, the regional water board members may further change the penalty amount determined by the staff.⁵

Discretion in enforcement and autonomy of the regional water boards are in part rooted in the Dickey Water Pollution Act of 1949. This act created the regional water boards, acknowledging that water pollution problems are regional—they are affected by rain and snowfall, the configuration of the land, and population density, as well as recreational, agricultural, urban, and industrial development. For this reason, the boundaries of the regional boards follow mountain chains and ridges that define watersheds, as opposed to political or administrative divisions, such as counties or cities. In addition, enforcement resources are different across regional water boards. For example, the average number of permits per compliance and enforcement staff during fiscal year 2006–7 varies from 109 in the Colorado River board to 539 in the Los Angeles board (table A4; tables A1–A13 are available online).

There have been concerns that the oversight of the regional boards by the state government might be insufficient, leading to inconsistencies in the implementation of the regulation across the different regions (Little Hoover Commission 2009). Related to these concerns, there have been

⁴ The list of the pollutants of groups I and II is in sec. 123.45 of Title 40 of the Code of Federal Regulations. Group I pollutants include biological oxygen demand and total suspended solids; examples of Group II pollutants are chlorine, copper, and cyanide.

⁵ For the penalty actions of 2009–15, we have data on the penalty amount assessed by staff as well as the final penalty amount. We find that the final penalty amount, as determined by the regional water board, differs from the staff assessment for 24% of the penalty actions. For more details, see app. A (apps. A–F are available online).

several reform proposals aiming at greater centralization of water quality regulation decisions at the state level. For example, in 2005, Assembly Bill 1727 would let the state board appoint the executive officer of each regional board. More drastically, the California Performance Review, a task force assembled in 2004 by the state government to restructure the state administration, proposed to abolish both the state and the regional boards and shift their environmental regulation duties to a new, centralized division of the California EPA (California Legislative Analyst's Office 2004).

D. Institutional Changes in Enforcement

In 1999, the Clean Water Enforcement and Pollution Prevention Act (SB 709) amended the California Water Code. To implement this new law, the state and regional water boards undertook various administrative efforts. Two of the major changes during our period of study are the launch of the California Integrated Water Quality System (CIWQS) in July 2005 and the establishment of the Office of Enforcement in July 2006.⁶ For convenience, we often refer to these two changes as the "2006 institutional changes."

These changes have decreased the administrative costs borne by the regional water boards in enforcing the water quality regulations. First, the new computer system, CIWQS, tracks and manages information about permittees, permits, inspections, violations, and enforcement activities. It also allows for the online submittal of self-monitoring data by permittees. Previously, dischargers would submit hard copies of the self-monitoring reports, which would then need to be manually entered into the system and reviewed by the staff. This system dramatically increased efficiency and enabled more resources to be devoted to enforcement.⁷ Second, the Office of Enforcement was established to provide statewide enforcement and to support the regional water boards' enforcement programs. The staff of the office regularly meet with representatives from the regional water boards to discuss enforcement matters and give feedback on enforcement approaches. Besides providing support to the regional water

⁶ In November 2006, Governor Schwarzenegger was reelected. The timing of these two administrative actions by the state government may be potentially related to the incumbent governor's reelection motives (List and Sturm 2006), but this is beyond the scope of our analysis.

⁷ The introduction of the computerized system did not lead to a decrease in the government budget for the water boards. Based on the historical budget publications, made available online by the state department of finance, the annual budget allocated for the support of the water boards regarding water quality issues (the item numbered as 3940-001-0001 until the 2008-9 budget and as 3940-001-0439 after) has been steady at around \$480 million in 2010 dollars.

boards, the office has the authority to perform independent enforcement actions.

E. Wastewater Treatment Facilities

We focus on the facilities that treat domestic wastewater and discharge the treated water. Based on our data, there are in total 288 such facilities that had an active NPDES permit during 2000–2014. They are responsible for the vast majority (73%) of effluent and water quality violations statewide during the period of study. A clear assessment of the compliance behavior of these facilities is thus particularly important for the better understanding of water pollution regulation in general.

Wastewater treatment facilities reduce the amount of oxygen-demanding substances, such as organic matter and ammonia; disinfect and chlorinate wastewater to decrease the concentration of infectious microorganisms; and remove phosphorus, nitrogen, and inorganic or synthetic organic chemicals. The process for treating wastewater includes a primary stage, in which solids are removed, and a secondary stage, which treats biological and dissolved organics. In addition, a tertiary stage may be used for disinfection and treatment of nitrogen, phosphorus, and other pollutants. The Clean Water Act requires municipal wastewater treatment plants to implement at least secondary-stage treatment.

A number of factors make it harder for some facilities to comply with the permit conditions than others. Facilities differ substantially in age, size, treatment technology, and capacity utilization rate (see table 1). They also differ in their finances. For example, although the state and federal governments provide subsidized financing to water treatment projects through the Clean Water State Revolving Fund, facilities located in small or disadvantaged communities often lack the resources and in-house expertise necessary to apply for grants and determine which types of projects are the most appropriate for their needs (California State Water Resources Control Board 2008). Moreover, weather conditions and existing pollution levels, which can substantially obstruct compliance efforts, vary both across facilities and over time.

III. Descriptive Statistics

A. Data Sources and Variables

We draw data from the CIWQS database for the universe of NPDES violations and enforcement actions during 2000–2014.⁸ The database provides

⁸ Data from before July 2005, when the water boards launched the CIWQS, were inputted retroactively into the CIWQS.

TABLE 1
SUMMARY STATISTICS: WASTEWATER TREATMENT FACILITIES

	BEFORE (2000–2005)		AFTER (2009–14)	
	Mean	Standard Deviation	Mean	Standard Deviation
Quarterly compliance and enforcement:				
Any effluent MMP violation	.23	.42	.17	.37
Number of effluent MMP violations	1.39	8.32	1.10	5.50
Any penalty upon effluent MMP violation	.59	.49	.77	.42
Penalty per effluent MMP violation (\$) ^a	2,249	4,721	2,624	4,641
Facility technological/cost attributes:				
Major facility	.75	.44	.75	.43
Design flow (millions of gallons per day)	17.83	52.32	19.24	54.88
First year of operation	1982.8	5.23	1982.9	5.30
1982–87	.69	.46	.70	.46
1988–	.06	.23	.05	.23
Advanced or tertiary treatment	.33	.47	.40	.49
Capacity utilization rate (%)	76.15	38.84	72.62	17.72
Over 87%	.26	.44	.24	.43
Resident population within the service area	94,858	297,458	157,321	476,886
Below 10,000	.36	.48	.35	.48
Run by a special district	.39	.49	.40	.49
High threat to water quality (by the board)	.53	.50	.55	.50
Time-varying quarterly local attributes:				
Precipitation (inches)	6.41	7.49	5.05	6.05
Below 25% of historic quarterly averages	.21	.41	.28	.45
Over 75% of historic quarterly averages	.28	.45	.22	.41
Average DO saturation (%)	10.83	12.92	12.62	13.44
Below 17%, swimmable	.71	.45	.55	.50
Time-invariant county attributes:				
Fresh water use for irrigation in 2010 (%)	.54	.32	.54	.32
Average household income in 2010 (\$ in thousands)	59,275	14,070	59,145	14,198
Population density per mile ² in 2010	764.99	1,754.8	762.87	1,788.8
Vote share for the 2006 water proposition (%)	50.21	8.94	50.24	9.06
Number of observations	6,159		6,620	
Number of facilities	228		215	

NOTE.—For the statistics on violations, we use the sample of 2002–5 for the period before the 2006 institutional changes and 2009–14 for the postchange period. We do not use the observations of 2000–2001 for analyzing the extent of compliance because we suspect that not all violation records are in the database for this early period. As for the penalty statistics, we employ the effluent MMP violations of 2000–2001 and 2009–10 and the follow-up penalty actions for 4 years. Acknowledging that the 2006 institutional changes may take time to be fully incorporated, we do not employ the observations of 2006–8. The key patterns found in the table are robust to our choice of the periods.

^a Average amount of total penalty per effluent or water quality MMP violation that occurred during 3 months, accounting for the penalties assessed within 4 years of the occurrence of the violation.

various attributes of each facility, which we include as control variables in our analyses: the EPA designation of the facility as a major facility;⁹ the water board designation of the extent of threat that the facility poses to the receiving water, ranging from category 1 to category 3;¹⁰ the starting year of its operation; and the unit of the local government that runs the facility (a county or city government or a special district). We collect further facility-level attributes from the Clean Watersheds Needs Survey data from the US EPA: the treatment level or technology (primary, secondary, or advanced); the extent of capacity utilization, defined by the ratio of the actual flow to the designed flow; and the size of population served by the facility.¹¹

We complement our data with the attributes of the county or the watershed in which the facility is located. As for the county-level attributes, we use various sources: the American Community Survey for average household income, the census for population and water use, and the California Irrigation Management Information System for precipitation. The precipitation data are provided at the 253 weather stations level, which we aggregate at the county level based on the stations' locations. From the California Secretary of State website, we obtain the vote shares for the 2006 Proposition 84 to authorize \$5.4 billion in bonds to fund various water projects. Last, we employ water pollution data from the US EPA Storage and Retrieval (STORET) and National Water Information System, provided by the US Geological Survey. We focus on the dissolved oxygen (DO) saturation of water, which represents the DO level divided by the maximum oxygen level given the water temperature. This is one of the most common omnibus measures of water quality because, among other reasons, it responds to a wide variety of pollutants (Keiser and Shapiro 2019). We closely follow the details and steps taken to clean the water pollution data in Keiser and Shapiro (2019) and construct the DO saturation measures for each facility by taking the average of water pollution readings within the area that shares the same five-digit hydrologic unit code as the facility during the previous 2 years. Following the same reference, we use the DO saturation cutoff of 17% to define surface water as "swimmable." Table 1 provides

⁹ For the NPDES program, the EPA designates certain facilities as major, depending on their industrial category or the amount of flow. Major facilities generally have flow greater than one million gallons per day or a discharge that poses a substantial threat to water quality.

¹⁰ Category 1 for the extent of threat to water implies that discharges of waste could cause the long-term loss of a designated beneficial use of the receiving water, such as the loss of drinking water supply and the closure of an area used for water contact recreation. The other two categories are less serious, either short term (category 2) or minor (category 3).

¹¹ Both treatment level and capacity utilization have been considered as important factors for compliance costs of wastewater treatment facilities in the engineering literature (Weirich, Silverstein, and Rajagopalan 2011; Suchetana, Rajagopalan, and Silverstein 2016; Rahm et al. 2018).

summary statistics of all variables mentioned above for the 228 wastewater treatment facilities used in our analysis.¹²

B. Scope of Analysis

We focus on effluent or water quality violations subject to the MMP of \$3,000. During the period of study, there are in total 48,155 violation records by domestic wastewater treatment facilities, and 19,740 (41%) of these records are categorized as MMP violations. Almost the entirety (99%) of the MMP records are associated with effluent or water quality violations.¹³ Some MMP violations are exempt from penalty by the State Water Code. We focus on nonexempt MMP violations only. As described in section II.B, MMP violations are either serious in the violation extent or chronic. Therefore, the violations that are not subject to the MMP regulation are relatively insignificant violations that were not repeated more than three times within 6 months.

C. Compliance and Enforcement over Time

Following the literature on water quality regulation (Magat and Viscusi 1990; Earnhart 2004a, 2004b; Shimshack and Ward 2005; Gray and Shimshack 2011), we assess compliance using the self-reported data. Two important features of the NPDES program ensure the reliability of the self-monitoring reports. First, the water boards conduct frequent inspections on the facilities. The inspection records show that 86% of the wastewater treatment facilities in our data received at least one inspection per year. Second, intentional misreporting can be punished by criminal sanctions to the responsible employees.¹⁴ If an employee has accurately reported operation conditions not in compliance with the NPDES permit, he or she cannot be held liable in a suit, because meeting the permit requirements is the responsibility of the permitted facility, not of that employee.

Table 1 shows that both compliance and enforcement stringency are higher after the institutional changes that we described in section II.D. The average number of MMP violations per quarter by a domestic wastewater treatment facility decreased from 1.39 in the 2002–5 period to 1.10

¹² Among the 288 domestic wastewater treatment facilities active during 2000–2014 in our data set, we exclude the following 60 from our analysis: (i) 48 facilities that were not matched to the Clean Watersheds Needs Survey Data, (ii) the only two facilities in region 6 (Lahontan), and (iii) 10 facilities owned by business, community organization, or the federal government.

¹³ The remaining 142 MMP violation records are regarding the timing of self-reports (126), order conditions (12), deficient monitoring (2), and enforcement actions (2).

¹⁴ According to sec. 122.22(d) of Title 40 of the Code of Federal Regulations, employees signing any report required by the permits must make a certification that they are aware of significant penalties for submitting false information, including the possibility of fines and imprisonment for misreporting violations.

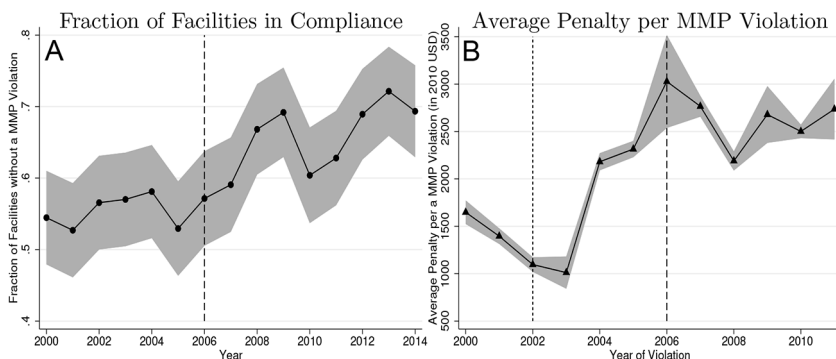


FIG. 1.—Compliance and enforcement. A, Fraction of the domestic wastewater treatment facilities without an effluent MMP violation for a given year. B, Average penalty per effluent MMP violation assessed within 4 years of the occurrence of the violation. Note that the 2006 institutional changes affected the within-4-year penalty for the violations that occurred in 2002 and after. Shaded areas represent the 95% confidence intervals.

in 2009–14, and the average penalties per MMP violation within 4 years of the violation's occurrence increased from \$2,249 in 2000–2001 to \$2,624 in 2009–10.¹⁵ The CIWQS database links each enforcement action with all associated violation records, which allows us to measure the enforcement stringency without having to make an assumption on the length of a lag before an enforcement action is taken.¹⁶

Figure 1 provides suggestive evidence that the aggregate changes in compliance and enforcement are associated with the institutional changes in 2006. First, figure 1A shows that the fraction of the facilities without an effluent MMP violation per year is relatively stable up until 2006, and then it substantially increases. Second, figure 1B shows that, focusing on penalties imposed within 4 years of the occurrence of a violation, the average penalty for a violation of 2006–11 (\$2,669) is greater than the counterpart for a violation of 2000–2002 by \$1,300, which is a 95% increase.¹⁷ For

¹⁵ Penalties may occur even after 4 years of the occurrence of a violation, but given the length of our panel data (15 years) and the usual length of a permit (5 years), we focus on the 4-year window. A large fraction of penalty actions occurs within 4 years—e.g., based on 937 effluent or water quality MMP violations that occurred in 2005 and were penalized by the end of 2014, the average lag before the first penalty record is 2.56 years, with a 90th percentile of 3.93 and a 99th percentile of 7.60 years.

¹⁶ When a penalty action is associated with multiple violation records, we divide the amount by the number of the linked violations to calculate the penalty amount for each individual record.

¹⁷ This increase in the average penalty is driven mainly by an increase in the probability of imposing penalty upon violation, as opposed to an increase in the amount of penalty conditional on any penalty actions. The probability of imposing penalty within 4 years of the occurrence of a MMP violation is 0.42 for a violation of 2000–2002, and this probability doubles for a violation of 2006–11. On the other hand, conditional on any penalty, the average penalties are similar across these two periods: 2000–2002 (\$3,280) and 2006–11 (\$3,211).

violations that occurred in 2003–5, the average penalty per violation is increasing, which is expected because the 2006 institutional changes were retroactive—a large fraction of the backlogged MMP violations that had not yet been penalized were penalized after the changes. Because the length of the period affected by the 2006 institutional changes within the 4-year penalty window increases as the violation date approaches 2006, we observe that the average penalty per violation increases during 2003–5 and then is constant from 2006 onward.

These changes may have been driven by time-varying factors, which may have coincided with the institutional changes. Table 1 shows that some facilities expanded their capacity (i.e., the average design flow increased) and/or adopted advanced or tertiary treatment technology after the institutional changes. Furthermore, table 1 documents that California became much drier during the period of study and that there was an increase in water pollution as measured by DO saturation levels.¹⁸ We find, however, that both compliance rates and penalties increased after the 2006 institutional changes, even after controlling for various time-varying facility and local attributes (table A1; fig. A1; figs. A1–A5 are available online).

D. Determinants of Penalty in the Data

Under the enforcement policy, as described in section II.B, the penalties for violations with the same attributes by different facilities may not be the same. California State Water Resources Control Board (2010) even recognizes that “the need to assess all of the applicable factors in liability determinations may yield different outcomes in cases that may have many similar facts” (10). There are three main reasons why different penalties may arise. First, the penalty amount is supposed to reflect the local beneficial use of the affected water, the violator’s ability to pay, and the economic benefit of the violation, all of which can differ across facilities. Second, the violator’s conduct is considered in penalty assessment so that the per-violation penalty may depend on the concurrent violations; that is, the penalty can be nonlinear in the number of violations. Third, the violator’s past compliance history is another factor in penalty assessment.

Table 2 shows the extent to which of these sources matter in practice, if any. First, we find that some facility and local characteristics further explain the penalty variation, even after controlling for violation attributes and current and past compliance behavior. For example, major facilities

¹⁸ According to the National Integrated Drought Information System, the longest drought in California since 2000 began in late December 2011 and ended on early March 2019, and the most intense period of drought occurred in October 2014. For more information on droughts in California, refer to <https://www.drought.gov/drought/states/california>.

TABLE 2
DETERMINANTS OF PENALTY FOR EFFLUENT MMP VIOLATIONS

	PENALIZED			LOG (PENALTY AMOUNT + 1)		
	(1)	(2)	(3)	(4)	(5)	(6)
Violation attributes:						
Priority violation ^a	.059 (.051)	.030 (.040)	-.001 (.033)	.506 (.398)	.292 (.314)	.032 (.262)
Group I pollutant	.066 (.050)	.059 (.038)	.049 (.030)	.489 (.383)	.416 (.302)	.336 (.237)
Group II pollutant	.040 (.074)	.030 (.062)	-.012 (.054)	.325 (.585)	.237 (.489)	-.074 (.427)
Current violations:						
Any violations (quarter)	.061 (.038)	.087** (.037)	.093*** (.033)	.599** (.291)	.779*** (.284)	.817*** (.256)
Past violations:						
Any violations (semester)	-.013 (.063)	-.003 (.055)	.047 (.044)	-.068 (.501)	.012 (.443)	.430 (.349)
Facility attributes:						
Major facility		.195*** (.065)	.103 (.065)		1.402*** (.519)	.692 (.499)
Started in 1982-87		.222*** (.086)	.167** (.068)		1.616** (.683)	1.267*** (.543)
Started in 1988-		.213 (.173)	.029 (.114)		1.492 (1.319)	.164 (.869)
Advanced/tertiary		-.036 (.072)	-.038 (.068)		-.381 (.566)	-.246 (.531)
Capacity utilization >87%		.005 (.079)	-.065 (.058)		-.064 (.634)	-.628 (.453)
Service population <10,000		-.057 (.069)	-.042 (.078)		-.545 (.534)	-.477 (.603)
Special district		.142** (.061)	.116** (.046)		1.014** (.489)	.817** (.369)
High threat		-.081 (.077)	-.002 (.078)		-.697 (.596)	-.097 (.618)
Weather pollution:						
Precipitation >75th percentile	-.130** (.050)	-.126*** (.042)	-.028 (.038)	-1.112*** (.396)	-1.080*** (.330)	-.248 (.274)
Precipitation <25th percentile	-.057 (.048)	-.038 (.036)	-.005 (.029)	-.402 (.379)	-.251 (.290)	-.0124 (.231)
Swimmable		.013 (.065)	.069 (.056)		.089 (.523)	.572 (.429)
County attributes:						
Irrigation water use >67%			.134* (.073)			1.119* (.583)
Household income >\$57,000			.158** (.076)			1.133* (.594)

TABLE 2 (Continued)

	PENALIZED			LOG(PENALTY AMOUNT + 1)		
	(1)	(2)	(3)	(4)	(5)	(6)
Population density >722/mi ²			.078 (.089)			.568 (.726)
Population density <80/mi ²			-.133* (.077)			-.990 (.603)
Proposition approval >50%			.118* (.063)			1.015** (.505)
Regional water board fixed effects	No	No	Yes	No	No	Yes
Year fixed effects; quarter fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Mean dependent variable	.678	.678	.678	5.239	5.239	5.239
Number of observations	15,827	15,827	15,827	15,827	15,827	15,827
Adjusted R ²	.167	.245	.400	.174	.245	.406

NOTE.—This table reports ordinary least squares estimates. The unit of observation is a violation; all nonexempt effluent MMP violations of 2000–2010 by the facilities in our sample are included. Standard errors are adjusted for clustering at the facility level and are provided in parentheses.

^a Water boards may categorize a violation as priority on the basis of the violation’s significance and severity (for the criteria for a priority violation, see California State Water Resources Control Board 2002, 2010).

* $p < .10$.
** $p < .05$.
*** $p < .01$.

and facilities located in a county with high average household income or high approval rate for the 2006 Proposition 84 are more likely to receive a penalty and tend to receive a larger penalty than other facilities. These patterns are possibly due to such facilities having relatively high marginal compliance costs. Alternatively, the regulator may consider violations by these facilities to be particularly serious, because of, for example, their larger potential harm to nearby residents or their potential political repercussions. The structural analysis developed in the next sections decomposes these distinct explanations.

Second, both the probability and the amount of penalty for a violation increase when there are other concurrent violations by the same facility. For example, according to column 3 in table 2, the probability of getting penalized within 4 years increases, on average, by 9.3 percentage points if there are other violations during the quarter when the given violation occurred. Moreover, the average penalty amount also increases significantly when there are other concurrent violations (cols. 4–6). In our facility-quarter-level analysis presented in section V, we find that the expected per-violation penalty increases as the number of violations during a quarter

increases.¹⁹ Our structural analysis allows us to study the effects of this nonlinearity of the penalty schedule on compliance.

Third, we do not find that past violations matter for penalty actions. In all specifications in table 2, the effects of having any other violation in the 6 months before a violation on penalty are statistically insignificant and also much smaller in magnitude than the counterpart effects of having any concurrent violations. Table A2 shows that these patterns persist even with an enriched set of variables based on current and past compliance behavior, following Blundell, Gowrisankaran, and Langer (2020). Granted, neither the Clean Water Act nor the California Water Code have an institutionalized dynamic deterrence mechanism, in contrast to the Clean Air Act (Blundell 2020; Blundell, Gowrisankaran, and Langer 2020). This motivates us to assume that the regulator sets a static penalty schedule in the analysis.²⁰

In appendix A, we provide evidence that these three empirical patterns are not necessarily driven by the violation attributes that are not controlled for. First, when we consider violations that are identical in terms of the pollutant, the emission standards, and the actual emission amounts, not only is there a large dispersion in penalties but also the three patterns described here persist. Second, focusing on penalty actions for which we observe both the initial penalty assessed by the staff and the final penalty set by the water board, we find that the penalty adjustment by the board varies with local attributes—in particular, water use for irrigation.

IV. Theoretical Model

A. Setup

In this section, we present our theoretical model of the interaction between a regulator and a single facility. Section V.A clarifies how we employ this model to analyze the heterogeneity in enforcement standards across different facilities. Our model is a version of the one developed by Mookherjee and Png (1994), with slight modifications to fit the empirical setting.²¹

¹⁹ We specify and estimate the penalty schedule as a function of the number of violations and various facility attributes (see eq. [13]). The penalty is allowed to be nonlinear in the number of violations, and our estimates provide evidence that the penalty is increasing and strictly convex in the number of violations (fig. 2; table A5).

²⁰ In the California Water Code, the MMP regulations have a partial feature of dynamic incentives in that, if four or more nonserious violations happen in six consecutive months, then the facility is subject to an MMP. Reflecting this, we employ 6 months as a unit of period instead of 3 months and find our results to be robust (see the results for alternative specification 3 in tables A12 and A13).

²¹ Departing from Mookherjee and Png (1994), we assume that the violations are stochastically determined by the facility's compliance efforts, which are not observed by the regulator. Also, we implicitly assume that exogenously determined inspection activities provide

Consider a wastewater treatment facility that chooses the extent to which it complies with regulations given a penalty schedule.²² We assume that the facility benefits from avoiding both compliance costs and penalty, following the evidence from section III.C that the facilities reduced the frequency of violations in response to an increase in penalty. The facility is better informed than the regulator about its compliance costs. Specifically, each facility is endowed with a type, θ , which is known to the facility only. The regulator knows that θ is the realization of a random variable Θ that follows a strictly increasing and continuously differentiable distribution function $F(\cdot)$ with support $(0, \bar{\theta})$. Let $f(\cdot)$ represent the associated density.

The facility sets a negligence level $a \in [0, \bar{a}]$, which is not observed by the regulator and affects the facility's compliance status in the following manner: let K be a random variable representing the number of emission violations incurred by the facility, and assume that K follows a Poisson distribution with mean a .²³ By setting the negligence level a , the facility derives private benefit $\theta b(a)$, which reflects the operation cost savings associated with lower compliance. Thus, denoting the benefit from maximum negligence or no compliance efforts as $\theta b(\bar{a})$, the total compliance cost by choosing some negligence level a is $\theta[b(\bar{a}) - b(a)]$. Because the facilities in our data are often publicly owned, $\theta b(a)$ could be different from the actual operational cost savings from emitting more pollutants to the waters, potentially reflecting the career concerns of the facility administrators and the scrutiny from the public.

Because the realization θ and the consequent negligence level a are not known by the regulator, a penalty schedule depends on the realized number of violations only. Specifically, given k violations, the facility has to pay the penalty according to a function $\epsilon(k)$. Assuming that the facility is risk neutral, we can restrict our attention to the expected penalty, conditional on a , which we denote by

$$e(a) \equiv \exp(-a) \sum_{k=0}^{\infty} \frac{\epsilon(k)}{k!} a^k. \quad (1)$$

The expected payoff to a facility setting the negligence level a is

incentives for facilities to truthfully self-report any violation, allowing us to abstract away from monitoring. The latter feature differentiates our analysis from those by Duflo et al. (2018) and Blundell, Gowrisankaran, and Langer (2020), which study the regulator's inspection actions and their associated costs.

²² We consider a continuous action for compliance, such as operation and maintenance of the existing pollution abatement equipment, as opposed to a discrete action, such as a large investment for a new piece of equipment. This assumption is suitable in our setting, given the discussion on the main causes of violation in app. A.

²³ The stochastic nature of the number of violations, given a negligence level, reflects that violations may occur because of circumstances outside of a facility's control, such as unexpected weather conditions and varying qualities of incoming water to the treatment facility.

$$-\theta[b(\bar{a}) - b(a)] - e(a). \quad (2)$$

We define that a negligence schedule, $a(\cdot)$, is implemented by a penalty schedule, $e(\cdot)$, if $a(\theta)$ maximizes (2) for all $\theta \in \Theta$. If $a(\cdot)$ is implemented by $e(\cdot)$, we have that

$$\theta b'[a(\theta)] = e'[a(\theta)] \quad (3)$$

whenever $a(\theta) > 0$. Given a penalty schedule $e(\cdot)$, the regulator's expected costs are

$$\int_0^{\bar{\theta}} (h[a(\theta)] + \psi e[a(\theta)] + \theta\{b(\bar{a}) - b[a(\theta)]\})f(\theta) d\theta, \quad (4)$$

where $\psi > 0$ denotes the marginal cost of imposing a penalty and $h(\cdot)$ represents the regulator's perceived environmental costs related to the facility emission violations.²⁴ The enforcement costs, given by $\psi e[a(\theta)]$, comprise the administrative and political costs associated with taking formal actions against a facility.²⁵

Lemma A1 in appendix B shows that by appropriately selecting the function $e(\cdot)$, the regulator is able to implement any continuous penalty schedule $e(\cdot)$ with a bounded domain. In the remainder of the paper, we define the regulator's problem as choosing $e(\cdot)$, subject to the constraint that the expected penalty for any a must be nonnegative and not exceed the facility's maximum amount of funds, ω :

$$0 \leq e(a) \leq \omega, \quad (5)$$

for any a .²⁶ The regulator's problem is to choose negligence and penalty schedules to minimize (4), subject to the constraints that the penalty schedule satisfies (5) and that it implements the negligence schedule.²⁷

²⁴ In our model, we assume that the regulator regards all violations by the same facility as identical to each other. In app. F, we reestimate the model, allowing for heterogeneity in the perceived harm of violations within a facility. Our key empirical findings persist.

²⁵ We assume that the marginal enforcement cost (ψ) is exogenous, i.e., not affected by the aggregate frequency of violations in equilibrium. The literature on crime, conversely, often treats these costs as endogenously determined by the policy maker's allocation of resources to law enforcement and the individuals' choices on criminal activities (see Fu and Wolpin 2018 and references therein). In environmental regulation, nonadministrative costs that might be out of the regulator's control, such as those associated with the political repercussions of enforcement actions, may explain a large part of the enforcement costs.

²⁶ Money raised from penalties is generally deposited in the Cleanup and Abatement Account, a fund managed by the state board from which the regional boards may request money for a project. Alternatively, publicly owned wastewater treatment facilities in small communities may be allowed to recover part of the amount they pay in penalties for compliance or supplemental environmental projects. Our analysis does not distinguish between these potential destinations of the penalties, since all facilities are liable to pay the penalty amount regardless. However, the possibility that some facilities are able to partially recover the penalties is one of the reasons why, in the empirical model presented in sec. V, the enforcement cost borne by the regulator may vary by facility attributes, such as the location of a facility.

²⁷ An individual rationality condition is not considered here. At optimum, $\lim_{\theta \rightarrow 0} e(\theta) = 0$, so the indirect (maximized) expected payoff for any $\theta \in (0, \bar{\theta})$ is nonnegative.

B. Characterization of Optimal Enforcement

We make the following assumptions on the facility's baseline compliance cost function, $b(\cdot)$, and the regulator's preference on water quality, $h(\cdot)$.

ASSUMPTION 1. $b(\cdot)$ and $h(\cdot)$ are continuously differentiable and strictly increasing.

Under assumption 1, the lemma in Mookherjee and Png (1994) shows that a negligence schedule, $a(\cdot)$, is implemented if and only if $a(\cdot)$ is non-decreasing and

$$\omega \geq \bar{\theta}b(\bar{a}) - \int_0^{\bar{\theta}} b[a(\theta)] d\theta. \quad (6)$$

The requisite expected penalty schedule is

$$e(a) = \theta(a)b(a) - \int_0^{\theta(a)} b[a(v)] dv, \quad (7)$$

where $\theta(a)$ denotes the highest type θ selecting an $a(\theta) \leq a$. Given this, the regulator's problem reduces to choosing $a(\cdot)$ to minimize

$$\int_0^{\bar{\theta}} \left(h[a(\theta)] + \psi \left[\theta b[a(\theta)] - \int_0^{\theta} b[a(v)] dv \right] + \theta \{ b(\bar{a}) - b[a(\theta)] \} \right) f(\theta) d\theta, \quad (8)$$

subject to $a(\cdot)$ being nondecreasing and (6). For simplicity, we assume that (6) is not binding at the optimum. By using integration by parts, we rewrite (8) as

$$\int_0^{\bar{\theta}} \left(h[a(\theta)] - \left[(1 - \psi)\theta + \frac{\psi[1 - F(\theta)]}{f(\theta)} \right] b[a(\theta)] + \theta b(\bar{a}) \right) f(\theta) d\theta.$$

We then consider point-wise optimization for each θ , and thus either $a(\theta) = 0$ or $a(\theta)$ satisfies the first-order condition

$$h'[a(\theta)] - b'[a(\theta)] \left((1 - \psi)\theta + \frac{\psi[1 - F(\theta)]}{f(\theta)} \right) = 0. \quad (9)$$

Proposition 1 shows that the following assumption, along with assumption 1, is sufficient to guarantee that the negligence schedule characterized above, denoted by $a^*(\cdot)$, is optimal and strictly increasing in θ for any θ such that $a^*(\theta) > 0$. See appendix B for the proof.

ASSUMPTION 2.

- i. $(1 - \psi)\theta + \psi[1 - F(\theta)]/f(\theta)$ is positive and strictly increasing in θ .
- ii. $b(\cdot)$ is strictly concave and $h(\cdot)$ is convex (or $b(\cdot)$ is concave and $h(\cdot)$ is strictly convex).

PROPOSITION 1. Under assumptions 1 and 2, the optimal negligence schedule, $a^*(\cdot)$, is continuous and nondecreasing in θ . For θ such that $a^*(\theta) > 0$, $a^*(\cdot)$ is characterized by (9) and strictly increasing in θ .

C. *Inefficiency in Enforcement*

For any $\theta \in (0, \bar{\theta})$, define the first-best level of negligence as $a^{**}(\theta) \equiv \max\{0, \tilde{a}^{**}(\theta)\}$, where $\tilde{a}^{**}(\theta)$ equates the marginal compliance costs and the marginal environmental costs:

$$h'[\tilde{a}^{**}(\theta)] - \theta b'[\tilde{a}^{**}(\theta)] = 0. \quad (10)$$

From (9), the equilibrium negligence level is not equal to this first-best level for two reasons: costly enforcement and the asymmetry of information between the regulator and the facility. Below we show that these two sources of inefficiency are deeply connected in the model; only in the presence of both does the equilibrium negligence level depart from the first best.

If there are no enforcement costs—that is, $\psi = 0$ —then it is optimal for the regulator to induce the facility to choose the first-best negligence level, by setting $e(a) = h(a) - h(0)$ for $a \in [0, \bar{a}]$, with large ω (specifically, $h(\bar{a}) - h(0) \leq \omega$). Intuitively, this schedule makes the facility fully bear the costs of externalities, which does not require the regulator's knowledge of its type θ .

Now suppose that the marginal enforcement cost is positive ($\psi > 0$) but the regulator can contract on the facility's type. That is, the regulator observes the realization θ of Θ and is able to set a penalty schedule conditional on this value. We maintain the assumption that the regulator observes the number of violations, k , but not the negligence level chosen by the facility, a . Then, for any given θ , it is optimal for the regulator to achieve the first-best negligence level by employing the following enforcement schedule:

$$e(a; \theta) = \begin{cases} 0 & \text{if } a \leq a^{**}(\theta), \\ \theta b'[a^{**}(\theta)]\{a - a^{**}(\theta)\} & \text{otherwise.} \end{cases}$$

This schedule is continuous in $a \in [0, \bar{a}]$, which guarantees its implementation by some $\epsilon(\cdot)$ by lemma A1 in appendix B and satisfies (5) with large ω (specifically, $\theta b'[a^{**}(\theta)]\{\bar{a} - a^{**}(\theta)\} \leq \omega$). It consists of a variant of a Pigouvian tax, which induces the facility to choose the first-best level of negligence, $a^{**}(\theta)$, with zero penalties (and thus no enforcement costs) in equilibrium.

V. Structural Model

A. Data-Generating Process

There are many facilities, which we index by i , and one regulator.²⁸ Periods are indexed by t . Let $\Theta_{i,t}$ represent the cost type for facility i in period t . For any given facility, the regulator sets the optimal penalty schedule, as described in section IV. Because of the potential changes in the primitives, the solution to the regulator's problem can also change over time. We denote by $e_{i,t}(\cdot)$ the penalty schedule for facility i in period t . Given $e_{i,t}(\cdot)$ and a realization of $\Theta_{i,t}$, the facility sets its optimal negligence level. As $\Theta_{i,t}$ is a random variable, the equilibrium negligence set by facility i in period t is also a random variable, which we denote by $A_{i,t}$. Let $G_{i,t}(\cdot)$ represent the distribution of negligence levels of facility i in period t .

The model primitives for facility i and period t are the distribution of facilities' types, $F_{i,t}(\cdot)$; the baseline compliance cost function, $b_{i,t}(\cdot)$; the regulator's perceived social cost of violations, $h_{i,t}(\cdot)$; and the marginal enforcement cost, $\psi_{i,t}$. The observables for facility i in period t are the number of violations, $K_{i,t}$; the amount of penalty; and facility-period characteristics, $\mathbf{x}_{i,t}$. The vector $\mathbf{x}_{i,t}$ includes (i) the facility's size, age, treatment technology, capacity utilization, and threat to water as categorized by the water board; (ii) the DO saturation level in the watershed (as a measure of the existing water pollution level) and the precipitation amounts; and (iii) the fraction of fresh water use for irrigation, the average household income, the population density, and the vote share for the 2006 Proposition 84 in the county where the facility is located (see table 5 or table A5 for a complete list). Each of these variables may affect all model primitives—for example, the population density of the facility's county is related to the size of population impacted by water pollution from the facility (and thus the social costs of violations), the nature of the incoming wastewater (which influences the compliance costs), and the administrative support for enforcement actions (which affects enforcement costs). Regional water board dummies may help account for heterogeneity of

²⁸ We consider a single generic regulator that deals with all facilities but allow that regulator to account for each facility's type distribution and to perceive the social cost of violations and the cost of enforcement differently for different facilities.

enforcement resources across the boards (see table A4), as well as other time-invariant unobserved local attributes.

We assume that facility heterogeneity in model primitives is captured by $\mathbf{x}_{i,t}$: specifically, $F_{i,t}(\cdot) = F_i(\cdot|\mathbf{x}_{i,t})$, $b_{i,t}(\cdot) = b_i(\cdot|\mathbf{x}_{i,t})$, $h_{i,t}(\cdot) = h_i(\cdot|\mathbf{x}_{i,t})$, and $\psi_{i,t} = \psi_i(\mathbf{x}_{i,t})$. Accordingly, the equilibrium penalty schedule and the equilibrium distribution of negligence levels are also functions of $\mathbf{x}_{i,t}$: $e_{i,t}(\cdot) = e_i(\cdot|\mathbf{x}_{i,t})$ and $G_{i,t}(\cdot) = G_i(\cdot|\mathbf{x}_{i,t})$. A key identifying assumption is that the primitives regarding facilities' compliance costs are time invariant—that is, $F_i(\cdot|\mathbf{x}_{i,t}) = F(\cdot|\mathbf{x}_{i,t})$ and $b_i(\cdot|\mathbf{x}_{i,t}) = b(\cdot|\mathbf{x}_{i,t})$. In particular, we assume that these primitives are unaffected by the 2006 institutional changes.²⁹ We further assume that $\Theta_{i,t}$ is independently distributed across facilities and periods, conditional on $\mathbf{x}_{i,t}$.³⁰ For ease of notation, we do not explicitly condition the model primitives on $\mathbf{x}_{i,t}$ in the discussion of identification below.

B. Identification

For the identification of the model, we follow three steps. First, we recover the distribution of negligence levels set by the facilities in each period, based on the observed violations. The second step, following the strategy proposed by d'Haultfoeulle and Février (2020), employs the exogenous change in the penalty schedule associated with the 2006 institutional changes to partially identify the facility type distribution and the marginal compliance cost function. This step does not rely on any assumption about the regulator's behavior.

The third step, which builds on the approach by Luo, Perrigne, and Vuong (2018), explores the restrictions imposed by the first-order conditions of the regulator to recover the social cost of negligence and the marginal enforcement cost, as well as to achieve exact identification of the type distribution and the marginal compliance cost function. By exploiting the exogenous variation in the penalty schedule, we are able to

²⁹ Our analysis controls for time-varying variables associated with the technology and cost structure of the facilities, such as the capacity utilization rate and the treatment level, and environmental factors including precipitation and water pollution. We argue that a systematic change in the compliance cost function beyond these observed attributes may not have occurred during the period of our study for two reasons. First, most facilities were operating well before and after 2006: 95% of the facilities in our sample started their operation before 1988, and the permit records show that the newest facility in our data set started its operation in 2004 and 13 facilities retired during 2000–2006. Second, based on the Census of Government Finance and Employment, there has been a steady flow of capital investment for sewerage services by local governments in 1997–2012, with an average of \$1.87 billion (in 2010 dollars) per year in total.

³⁰ By this assumption, each facility independently draws its type every period. An alternative assumption is that the facilities' types are constant over time and the regulator commits not to exploit the information on the facility type obtained in the previous periods. Our identification argument holds under either of these two assumptions, which are both consistent with the static penalty schedules in the data (tables 2, A2, A3).

consider a more flexible form for the regulator's objective function than would be possible using their approach alone. In particular, they assume that $h_i(\cdot)$ (the monopolist's cost function in their setting) is linear, whereas we can accommodate a polynomial specification of arbitrary degree.

We restrict our attention to the case in which it is optimal for all facilities to choose a nonzero rate of violations, or $a_t(\theta) > 0$, for any period t and $\theta \in (0, \bar{\theta})$.³¹ Then, given any period t , the distribution of the number of violations by any facility is a mixture Poisson. Indeed, a facility chosen at random sets a negligence level according to the distribution $G_t(\cdot)$, and given the negligence level, the number of violations for that facility follows a Poisson distribution. The following lemma establishes the identification of $G_t(\cdot)$ from the observed number of violations across facilities. To prove this lemma, we exploit the moment-generating function of the Poisson distribution, which was also used in Aryal, Perrigne, and Vuong (2019). See appendix B for the proofs of the lemmas and the propositions in this section.

LEMMA 1. For every t , $G_t(\cdot)$ is identified.

Having identified the distribution of negligence levels in each period, our strategy to partially identify $b(\cdot)$ and $F(\cdot)$ closely follows that proposed by d'Haultfoeuille and Février (2020). We consider two enforcement regimes, before and after the 2006 institutional changes, and assume that within each regime, the penalty schedule does not change. Formally, we make the following assumption on $e_t(\cdot)$, the expected penalty in period t , as a function of the negligence level set by the facilities:

ASSUMPTION 3. $e_t(\cdot) = e_{\text{pre}}(\cdot)$ for all $t < 2006$. Similarly, $e_t(\cdot) = e_{\text{post}}(\cdot)$ for all $t > 2008$. Moreover, $e'_{\text{post}}(a) > e'_{\text{pre}}(a)$ for all $a > 0$.

We can directly identify the penalty schedule, $e_t(\cdot)$ as a function of the number of violations, in each period. From (1), therefore, we readily identify the functions $e_{\text{pre}}(\cdot)$ and $e_{\text{post}}(\cdot)$, so assumption 3 is testable.³² The latter part of the assumption implies that the enforcement regime becomes stricter after the institutional changes. We exclude the period of 2006–8 as a transition period, although such an exclusion is not necessary and the length of the transition period can be adjusted.

Under assumption 3, any facility of a given type θ sets at most two different negligence levels—one for each of the two enforcement regimes.

³¹ This assumption is consistent with our data—where 10% of the facilities that were active during all the 60 quarters in 2000–2014 were always in compliance—in the sense that a very small but positive value of a can generate no violations for a long period of time. For example, a facility that sets $a = 0.001$ will have no violations during 60 periods with probability $e^{-0.001 \times 60} \approx 0.94$.

³² Given that we assume that $F(\cdot)$ and $b(\cdot)$ remain the same for the whole sample period, assumption 3 can be related to the changes in $h_i(\cdot)$ and ψ_i due to the institutional changes. Specifically, a sufficient condition for this assumption to hold is that $h'_{\text{post}} \geq h'_{\text{pre}}(\cdot)$ and $\psi_{\text{post}} \leq \psi_{\text{pre}}$, with at least one inequality being strict. See app. B for comparative statics of the model.

Accordingly, we denote by $G_j(\cdot)$ the distribution of negligence levels holding in period $j \in \{\text{pre}, \text{post}\}$, where, as above, “pre” refers to $t < 2006$ and “post” refers to $t > 2008$. Also, we denote by $\tilde{a}(\cdot, j)$ the equilibrium negligence function in period $j \in \{\text{pre}, \text{post}\}$. From (3), it is clear that $\tilde{a}(\theta, \text{pre}) > \tilde{a}(\theta, \text{post})$ for all θ . Let the supports of the negligence level distributions before and after the regime change be given by $\mathcal{A}_{\text{pre}}$ and $\mathcal{A}_{\text{post}}$, respectively. We assume that $\mathcal{A}_{\text{pre}} \cap \mathcal{A}_{\text{post}} \neq \emptyset$.

The strategy described below and formalized in proposition 2 allows us to partially recover $F(\cdot)$ and $b(\cdot)$ without making any assumptions about the behavior of the regulator. Define the function $\tilde{\theta}(a, j)$ as the inverse of $\tilde{a}(\cdot, j)$ for any $a \in \mathcal{A}_j$. Define also the following two functions:

$$T^H(a) \equiv G_{\text{pre}}^{-1}[G_{\text{post}}(a)], \quad (11)$$

$$T^V(\theta, a) \equiv \frac{e'_{\text{post}}(a)}{e'_{\text{pre}}(a)} \theta. \quad (12)$$

The function $T^H(\cdot)$ is defined for any $a \in \mathcal{A}_{\text{pre}} \cap \mathcal{A}_{\text{post}}$, while $T^V(\cdot, \cdot)$ is identified over the entire domain of a and θ . The following lemma plays a key role in the identification of $F(\cdot)$ and $b(\cdot)$:

LEMMA 2. Under assumptions 1–3, we have that $T^H(a) = \tilde{a}[\tilde{\theta}(a, \text{post}), \text{pre}]$ for $a \in \mathcal{A}_{\text{pre}} \cap \mathcal{A}_{\text{post}}$ and $T^V[\tilde{\theta}(a, \text{pre}), a] = \tilde{\theta}(a, \text{post})$ for any $a \in \mathcal{A}_{\text{pre}}$.

This lemma establishes that $T^H(a)$ returns the negligence exerted in the pre regime by a facility type that, while in the post regime, exerted negligence level a ; and $T^V[\tilde{\theta}(a, \text{pre}), a]$ returns the type that exerts negligence level a in the post regime.

To partially identify $F(\cdot)$ and $b(\cdot)$, we normalize $\tilde{\theta}(a_0, \text{post}) = \theta_0 = 1$ for some $a_0 \in \mathcal{A}_{\text{post}}$.³³ We then define recursively

$$a_l = T^H(a_{l-1})$$

and

$$\theta_l = T^V(\theta_{l-1}, a_l).$$

The transform $T^H(\cdot)$ connects points in the negligence distribution supports in both regimes. For any $a \in \mathcal{A}_{\text{post}}$, $T^H(a) \in \mathcal{A}_{\text{pre}}$. However, under assumption 3, we have that $T^V[\tilde{\theta}(a, \text{pre}), a] > \tilde{\theta}$ for $a > \max(\mathcal{A}_{\text{post}})$; that is, there are relatively high negligence levels that, in equilibrium, are set only in the pre regime. Let $\bar{L}$ represent the largest integer such that $T^H(a_{\bar{L}}) \in \mathcal{A}_{\text{post}}$. We are now ready to state the following result.

³³ This location normalization is necessary, as only the product of the cost type and the baseline function enter the facility's objective function. Thus, multiplying the type by a constant and dividing the baseline function by the same constant results in an observationally equivalent model structure. In our empirical application, we set $a_0 = 1$.

PROPOSITION 2. Suppose that assumptions 1–3 hold. Then, for any $l \in \{0, 1, \dots, \bar{L}\}$ and $j \in \{\text{pre}, \text{post}\}$, the following objects are identified up to the normalization $\tilde{\theta}(a_0, \text{post}) = \theta_0 = 1$ for some $a_0 \in \mathcal{A}_{\text{post}}$: (i) the equilibrium negligence level, $\tilde{a}(\theta_l, j)$; (ii) the distribution of cost types, $F(\theta_l)$; and (iii) the marginal baseline compliance cost function, $b'[\tilde{a}(\theta_l, j)]$.

The proof, which we show in appendix B, is based on d'Haultfoeuille and Février (2020). Here we simply outline it. Starting from the normalization $\tilde{\theta}(a_0, \text{post}) = 1$, we apply the transforms $T^H(\cdot)$ and $T^V(\cdot)$ in an iterative fashion to recover a sequence of types $\{\theta_l\}_{l \in \{0, \dots, \bar{L}\}}$, together with the corresponding negligence levels $\tilde{a}(\theta_l, \text{pre})$ and $\tilde{a}(\theta_l, \text{post})$ set by each θ_l in equilibrium in the pre- and post-2006 periods, respectively. For every identified pair θ_l and $\tilde{a}(\theta_l, j)$, we use (3), the first-order condition of the facility's problem, to obtain $b'[\tilde{a}(\theta_l, j)] = e'_j[\tilde{a}(\theta_l, j)]/\theta_l$. Finally, because of the one-to-one mapping between types and equilibrium negligence levels (proposition 1), we can recover $F(\theta_l) = G_j[\tilde{a}(\theta_l, j)]$. Under the assumptions of proposition 2, $F(\cdot)$ and $b'(\cdot)$ are identified only over a finite set of values. The set is finite due to the boundedness of the type space, and the exact number of values at which the functions are identified depends on the shape of the functions $\tilde{a}(\cdot, \text{pre})$ and $\tilde{a}(\cdot, \text{post})$.

To complete the identification of the model, we must explicitly consider the regulator's problem. We begin by making the following simplifying assumption.

ASSUMPTION 4.

- i. $h_t(\cdot) = h_{\text{pre}}(\cdot)$ and $\psi_t = \psi_{\text{pre}}$ for all $t < 2006$, and $h_t(\cdot) = h_{\text{post}}(\cdot)$ and $\psi_t = \psi_{\text{post}}$ for all $t > 2008$.
- ii. For $j \in \{\text{pre}, \text{post}\}$, the function $h_j(a)$ is a polynomial function of a finite degree R with $h_j(0) = 0$; that is, $h_j(a) = \sum_{r=1}^R \gamma_{j,r} a^r$ for any R .

Assumption 4(i) implies that all model primitives are constant within each of the two regimes, and assumption 4(ii) imposes a flexible parametric structure to the external costs of violations as perceived by the regulator. Below we make a technical assumption on the equilibrium penalty schedule, which guarantees that we can employ the first-order conditions from the regulator's problem to recover ψ_j and $\gamma_{j,r}$ for $j \in \{\text{pre}, \text{post}\}$ and $r \in \{1, \dots, R\}$; then we state our last proposition.

ASSUMPTION 5. There is an interval $U \in \mathbb{R}_+$ such that the functions $\tilde{E}_0(a) \equiv e'_{\text{post}}(a)/e'_{\text{pre}}(a)$ and $\tilde{E}_{j,r}(a) \equiv a^r/e'_j(a)$ for all $r \in \{1, \dots, R\}$ are strictly monotone in $a \in U$.

PROPOSITION 3. Suppose that assumptions 1–5 hold. Then, if $\bar{L} \geq 1$, the following objects are identified up to the normalization $\tilde{\theta}(a_0, \text{post}) = 1$ for some $a_0 \in \mathcal{A}_{\text{post}}$: (i) the distribution of facilities' types, $F(\cdot)$; (ii) the derivative of the baseline compliance cost function, $b'(a)$, for any $a \in \mathcal{A}_{\text{pre}} \cup \mathcal{A}_{\text{post}}$;

and (iii) the parameters of the regulator's objective function, $\{\gamma_{j,r}\}_{r=1}^R$ and ψ_j , for $j \in \{\text{pre}, \text{post}\}$.

In a nutshell, we first identify the parameters of $h_j(\cdot)$ and ψ_j based on (9), the first-order condition of the regulator, evaluated at the vector $\{\theta_l\}_{l=0}^L$ for which $\tilde{a}(\theta_l, \text{pre})$ and $\tilde{a}(\theta_l, \text{post})$ are known from proposition 2. The main challenge in the process is that $f(\theta_l)$ is not yet identified. To address the challenge, we exploit the relationship between a density and its quantile function, a technique that has been employed by Luo, Perrigne, and Vuong (2018). Equation (9) holds for every θ in the support—not only those in $\{\theta_l\}_{l=0}^L$. Thus, once $h_j(\cdot)$ and ψ_j are identified, we can use (9) to identify $\tilde{a}(\theta, j)$ for all θ . Knowing the whole mapping $\tilde{a}(\cdot, j)$, we employ the same techniques explained in the discussion of proposition 2 to recover $F(\cdot)$ and $\mathcal{V}(\cdot)$.

C. Estimation

We first use flexible parametric functional forms to estimate the penalty schedules and the distribution of negligence levels before and after the 2006 institutional changes. Although these objects can be nonparametrically estimated in principle, our sample size and our intent to condition the estimates on $\mathbf{x}_{i,j}$ render such an approach infeasible. We then closely follow our identification strategy to estimate the model primitives. We describe the procedure below and in appendix C.

1. Estimation of the Penalty Schedules

We assume that the penalties, denoted by $\epsilon_{i,t}$, take a type II Tobit form. Recall that $k_{i,t}$ denotes the number of violations by facility i in period (quarter in our estimation) t , and define

$$\begin{aligned}\epsilon_{1i,t}^* &= \mathbf{x}_{i,t}\phi_{1,x} + \mathbf{1}_{\{t > 2006\}}\phi_{1,\text{post}} + \phi_{1,k}k_{i,t} + u_{1i,t}, \\ \log \epsilon_{2i,t}^* &= \log[\exp(\mathbf{x}_{i,t}\phi_{2,x})k_{i,t} + \phi_{2,k^2}k_{i,t}^2] + u_{2i,t}, \\ \text{and } \epsilon_{i,t} &= \begin{cases} \epsilon_{2i,t}^* & \text{if } \epsilon_{1i,t}^* \geq 0, \\ 0 & \text{otherwise,} \end{cases}\end{aligned}\tag{13}$$

where $(u_{1i,t}, u_{2i,t})$'s are i.i.d. draws from a bivariate normal distribution with zero mean, variances 1 and σ_2^2 , and covariance σ_{12} . We assume that $(u_{1i,t}, u_{2i,t})$ is independent of $(k_{i,t}, \mathbf{x}_{i,t})$ and that these error terms represent features of the facility and the occurred violations that might affect penalties ex post.³⁴

³⁴ One important dimension that $(u_{1i,t}, u_{2i,t})$ might capture is violation severity. One could expect the number and severity of occurred violations to be related in reality, making

This econometric specification of the observed penalties closely reflects the institutional features from section II. First, not all MMP violations are penalized, and both the event of nonzero penalty and the penalty amount depend on violations and facility attributes. Second, the penalty amount can be nonlinear in the number of violations. Third, to account for the institutional changes in 2006, we allow the probability of a nonzero penalty to depend on a dummy variable indicating the post-2006 period.³⁵

Denote $\phi \equiv \{\phi_{1,x}, \phi_{1,\text{post}}, \phi_{1,k}, \phi_{2,x}, \phi_{2,k^2}, \sigma_2, \sigma_{12}\}$. We estimate ϕ using the facility-quarter observations during 2000–2001 and 2009–10 with at least one MMP violation, while truncating observations with an extreme number of violations—more than 25 (the 95th percentile conditional on at least one MMP violation).³⁶ Let $\epsilon_{\text{pre}}(k|\mathbf{x}_{i,t}; \phi)$ and $\epsilon_{\text{post}}(k|\mathbf{x}_{i,t}; \phi)$ denote the expected penalties for a facility with characteristics $\mathbf{x}_{i,t}$ and k violations in a quarter before and after the 2006 institutional changes, respectively. Given our Tobit model, these expected penalties can be written as

$$\epsilon_j(k|\mathbf{x}_{i,t}; \phi) = \Pr(\epsilon_{1i,t}^* \geq 0 | k, \mathbf{x}_{i,t}, \phi) \mathbb{E}[\epsilon_{2i,t}^* | \epsilon_{1i,t}^* \geq 0, k, \mathbf{x}_{i,t}, \phi].$$

We obtain an estimate $\hat{\phi}$ by maximum likelihood estimation (MLE) and substitute $\hat{\phi}$ for ϕ in the above equation. Then, using (1), we estimate $e_{\text{pre}}(a|\mathbf{x}_{i,t})$ and $e_{\text{post}}(a|\mathbf{x}_{i,t})$, the penalty schedules for a facility with attributes $\mathbf{x}_{i,t}$, as functions of its negligence level a . To do so, we approximate the infinite sum in (1) by a finite sum up to a large number as follows:

$$\hat{e}_j(a|\mathbf{x}_{i,t}) = \exp(-a) \sum_{k=0}^{k=150} \frac{\epsilon_j(k|\mathbf{x}_{i,t}; \hat{\phi})}{k!} a^k. \quad (14)$$

2. Estimation of the Negligence Distribution

We assume that the negligence level distribution, $G_j(\cdot|\mathbf{x}_{i,t})$ for $j \in \{\text{pre}, \text{post}\}$, is a gamma distribution with shape parameter $\exp(\mathbf{z}_{i,t}\delta)$ and rate parameter $\exp(\mathbf{z}_{i,t}\delta - \beta_{0,j} - \beta_1\mathbf{x}_{i,t})$, where the vector $\mathbf{z}_{i,t}$ is a subset of $\mathbf{x}_{i,t}$.³⁷ Consistent with the theoretical model, conditional on the realization of

our assumption of independence between $(u_{1i,t}, u_{2i,t})$ and $(x_{i,t}, k_{i,t})$ implausible. In app. F, we consider an extension of the model that explicitly incorporates violation severity as an endogenous variable. The results obtained from the extended model are very similar to those presented in the main text.

³⁵ In a more general specification, we allow the 2006 institutional changes to affect both the probability and the magnitude of nonzero penalties. We find that the latter effect is statistically insignificant. This is consistent with the empirical patterns of the penalty changes over time (see n. 17).

³⁶ Our findings are robust to reasonable perturbations of the truncation criterion. See the results for alternative specifications 4 and 5 in tables A12 and A13.

³⁷ We include in $\mathbf{z}_{i,t}$ region-specific dummies and an intercept. Alternative specification 6 in tables A12 and A13 show that the results are robust to including more variables in $\mathbf{z}_{i,t}$.

a negligence level a , the distribution of violations is Poisson with mean a . Define $\beta_j \equiv (\beta_{0,j}, \beta_1)$ for $j \in \{\text{pre}, \text{post}\}$. Then the estimation of the distribution of negligence levels amounts to estimating the parameter vectors δ , β_{pre} , and β_{post} . We estimate these parameters by MLE, using the observed $k_{i,t}$ and $\mathbf{x}_{i,t}$.

3. Estimation of the Model Primitives

The remainder of the estimation procedure, which closely follows the identification strategy in section V.B, only takes as an input the penalty schedules and the negligence distributions, before and after 2006. Thus, having estimated these objects using data from all facility periods, we estimate the model primitives separately for any observed attributes $\mathbf{x}_{i,t}$. In our empirical analysis, we constrain the social costs of negligence as perceived by the regulator to be linear. That is, we assume that $h_j(a|\mathbf{x}_{i,t}) = \gamma_j(\mathbf{x}_{i,t})a$ for any a and $j \in \{\text{pre}, \text{post}\}$. This constraint facilitates the interpretation of our empirical results, as it keeps the social costs of negligence for each facility one-dimensional—making the comparison of costs across different facilities convenient. As discussed in section VI, it also allows for a good fit of the model to the data. In sum, the estimated model primitives for any given $\mathbf{x}_{i,t}$ are the functions $F(\cdot|\mathbf{x}_{i,t})$ and $b'(\cdot|\mathbf{x}_{i,t})$, characterizing the distribution of facility compliance costs, and the regulator preferences parameters before and after the institutional changes, $\gamma_{\text{pre}}(\mathbf{x}_{i,t})$, $\gamma_{\text{post}}(\mathbf{x}_{i,t})$, $\psi_{\text{pre}}(\mathbf{x}_{i,t})$, and $\psi_{\text{post}}(\mathbf{x}_{i,t})$.

VI. Results

Given that the estimated model primitives are functions of observed attributes ($\mathbf{x}_{i,t}$), we obtain each i th facility's primitives based on $\mathbf{x}_{i,t}$ in the first quarter of 2005 separately for all 221 facilities that were active during that quarter and build our results. We provide the summary statistics of the results across the facilities and bootstrap standard errors for the model fit and counterfactual results.

A. Model Fit

We find that the estimated model fits the data well. Table 3 compares the distributions of the number of quarterly violations and average quarterly penalty, as predicted by the estimated model, with the counterpart distributions observed in the data. The estimated model is able to reproduce both the high probability of no violations at the facility-quarter level and the shift in the distribution of violations and penalties that took place following the 2006 changes. In addition, figure 2 shows the fit for the point-wise average (per-violation) penalty for each number of violations across

TABLE 3
MODEL FIT

	BEFORE		AFTER	
	Data	Model	Data	Model
Number of violations:				
0	.771	.789 [.744, .800]	.843	.805 [.793, .816]
1	.073	.063 [.061, .068]	.045	.062 [.060, .067]
2	.038	.031 [.030, .033]	.027	.030 [.029, .032]
3	.029	.020 [.019, .021]	.019	.019 [.018, .020]
4	.017	.014 [.013, .015]	.013	.013 [.012, .014]
5 or more	.072	.082 [.075, .091]	.053	.071 [.063, .077]
Penalties:				
Average penalty (\$)	2,381	2,318 [1,833, 3,158]	2,215	2,119 [1,812, 2,704]

NOTE.—This table provides the estimated distributions of the number of violations and penalty amounts across all facilities, as observed in the data and predicted by the fitted model. For the distribution of the number of violations, the data consist of facility-quarter observations in the periods 2002–5 and 2011–14. For the penalty statistics, we use facility-quarter observations in the periods 2000–2001 and 2009–10. Bootstrap 95% confidence intervals for the fitted values are given in brackets.

facilities based on the first-stage estimates. The estimated penalty schedule exhibits strict convexity: the average per-violation penalty is strictly increasing in the number of violations (panel B). This is consistent with our violation-level descriptive findings (tables 2, A3).

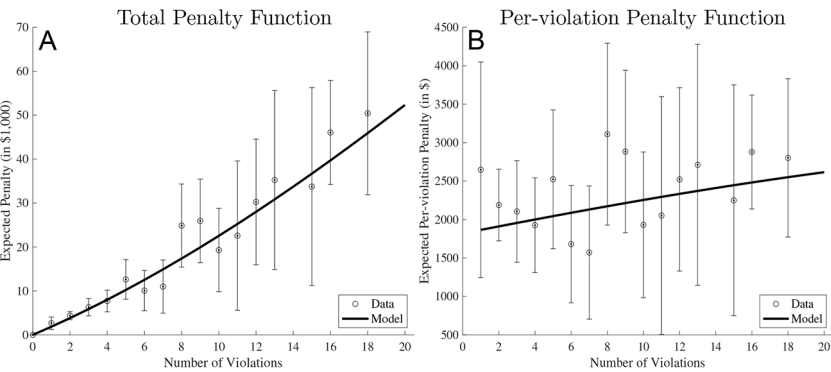


FIG. 2.—Model fit: penalty as a function of violations. A, Average penalties conditional on each number of violations during a quarterly period in the data and as predicted by the first-stage estimates, respectively, before the 2006 institutional changes. The 95% confidence intervals for the average penalties in the data are indicated by the error bars. B, Average amount of per-violation penalties.

TABLE 4
MODEL PRIMITIVE ESTIMATES: SUMMARY STATISTICS

	Median	Interquartile Range
Facility primitives:		
Marginal compliance cost	1,431.0 [1,072.1, 1,714.8]	1,084.4 [844.8, 1,619.1]
Regulator primitives:		
Marginal external/social cost of violation (γ):		
Before the 2006 changes	3,589.4 [2,892.7, 4,547.9]	1,603.1 [1,002.8, 2,812.8]
After the 2006 changes	3,157.1 [2,771.9, 3,998.6]	912.8 [765.1, 1,742.2]
Difference before and after the 2006 changes	-114.5 [-632.5, 103.9]	1,013.8 [317.9, 1,767.6]
Marginal enforcement cost (ψ):		
Before the 2006 changes	.865 [.361, .999]	.482 [.000, .701]
After the 2006 changes	.204 [.085, .491]	.219 [.114, .353]
Difference before and after the 2006 changes	-.551 [-.745, -.239]	.328 [.166, .542]

NOTE.—This table provides the summary statistics of the marginal compliance cost (see n. 39) and the regulator preference parameters (γ , ψ), before and after the 2006 institutional changes. Bootstrap 95% confidence intervals are given in brackets.

In appendix E, we show that the estimated model satisfies the assumptions in sections IV and V. Recall that assumptions 1 and 2 are sufficient for the equilibrium characterization and assumptions 3–5 are sufficient for the identification of the model.

B. Estimated Model Primitives

Table 4 presents the summary statistics of our estimates of the primitives of the model: the facilities’ marginal compliance costs and regulator preferences (γ and ψ) for the periods before and after the 2006 institutional changes.³⁸ Noting that our model primitives include the marginal compliance cost function and the cost type distribution, we report the median marginal compliance costs evaluated at the facility’s median value of θ and the median value of the negligence level before the 2006 institutional changes, averaged across all facilities.³⁹

Our estimates in table 4 imply that the increase in enforcement stringency after the 2006 institutional changes, as documented in table 1 and figure 1, is rationalized by a decrease of the marginal enforcement costs

³⁸ Table A5 provides the penalty schedule estimates (ϕ ’s) and the negligence distribution parameter estimates (δ and β ’s) described in secs. V.C.1 and V.C.2. Table A6 provides summary statistics of the regulator preference estimates for each regional board.

³⁹ We calculate the marginal compliance cost of a given facility with covariates $\mathbf{x}$, $\hat{\theta}b'(a|\mathbf{x})$, evaluated at $\theta = \hat{F}^{-1}(0.5|\mathbf{x})$ and $a = (1/n)\sum_i \hat{a}_{pre}^*[\hat{F}^{-1}(0.5|\mathbf{x}), \mathbf{x}] = 0.01$, for the summary statistics in table 4.

(ψ). The median value of the enforcement cost per extra dollar of penalty is \$0.86 before the 2006 changes and then falls by \$0.66. This decrease is both economically and statistically significant; however, the changes in the marginal external cost of violation as perceived by the regulator (γ) are statistically insignificant. These findings seem to be consistent with the nature of the institutional changes that aimed to reduce the administrative burden of imposing penalties borne by each regulator in the regional water boards, by providing the computerized information system and the support from the Office of Enforcement.

C. *Regulator vs. Local Resident Preferences*

We find suggestive evidence that the estimated regulator preferences on compliance at least partially represent those of the population at the facility location. Columns 2 and 3 of table 5 provide the results of regressing the logarithm of the regulator preference parameter estimates for the periods before the 2006 institutional changes, $\gamma_{\text{pre}}(\mathbf{x}_{i,t})$, and $\psi_{\text{pre}}(\mathbf{x}_{i,t})$, respectively, on all facility and local attributes used in the estimation.⁴⁰ First, the regulator's perceived external cost of violation, $\gamma_{\text{pre}}(\mathbf{x}_{i,t})$, is 47% higher for a facility located in a county with a high average household income (over \$57,000 in 2010) than a facility located in a poorer county (col. 2 of table 5), and the enforcement costs per extra dollar of penalty, $\psi_{\text{pre}}(\mathbf{x}_{i,t})$, are 29% lower (col. 3). Given that environmental goods or amenities are in general considered a normal good, we view this finding as a supporting piece of evidence that regulators reflect local preferences on water quality in their enforcement decisions. Second, we also find that the marginal enforcement costs are 13% lower in counties where the majority of the voters supported the 2006 Proposition 84.

These findings may be driven by the state government's political considerations, which might partially determine the allocation of enforcement resources, and the regional board members' ability and willingness to tailor the enforcement standards to local preferences and needs. Because regional board members are paid by the hour at a relatively low rate while their job requires significant expertise, we speculate that they are likely to serve the boards out of civic duty or personal political aspirations, which may help align their actions with the local constituents' preferences.

D. *Regulator Preferences and Penalty Disparities*

Figure 3A provides the scatterplot of (i) the expected penalties for each facility, evaluated at the average negligence level before the 2006 changes

⁴⁰ The regression results on the estimated regulator preferences for the periods after the 2006 changes are presented in table A7 and are qualitatively similar to those presented here.

TABLE 5
EXPLAINING COMPLIANCE COSTS AND REGULATORS' PREFERENCES

	Log(Compliance Cost) (1)	Log γ (2)	Log ψ (3)
Facility attributes:			
Major facility	.452*** (.100) [.421]	.149*** (.038) [.136]	-.110 (.068) [.124]
Started in 1982-87	-.309*** (.108) [.361]	.025 (.032) [.098]	.031 (.065) [.107]
Started in 1988-	-2.65*** (.486) [.529]	-.099* (.051) [.097]	-.311* (.168) [.120]
Advanced/tertiary	.057 (.089) [.313]	-.076*** (.029) [.099]	-.135*** (.055) [.081]
Utilization >87%	.110 (.089) [.462]	-.055* (.029) [.103]	.097 (.055) [.081]
Service population <10,000	.125 (.094) [.307]	.048 (.042) [.104]	-.109* (.056) [.123]
Special district	-.404*** (.074) [.832]	-.292*** (.026) [.192]	.030 (.046) [.345]
High threat	.208** (.088) [.403]	-.001 (.035) [.083]	-.189*** (.054) [.112]
Weather and pollution:			
Precipitation >75th percentile	-.223 (.177) [.564]	-.069** (.034) [.132]	.048 (.056) [.149]
Precipitation <25th percentile	.440*** (.192) [.264]	.072 (.094) [.108]	.003 (.170) [.142]
Swimmable	-.127* (.066) [.568]	-.249*** (.055) [.092]	-.159* (.087) [.191]
County attributes:			
Irrigation >67%	-.274** (.132) [4.649]	-.004 (.036) [.288]	.012 (.079) [.342]
Income >\$57,000	.776*** (.118) [.576]	.469*** (.043) [.194]	-.291*** (.077) [.223]
Density >722/mi ²	-.631*** (.182) [.951]	-.267*** (.037) [.182]	.039 (.080) [.212]
Density <80/mi ²	.172* (.099) [.482]	.083* (.043) [.169]	.019 (.080) [.224]
Proposition >50%	.183 (.127) [.302]	-.009 (.037) [.145]	-.134* (.078) [.150]

TABLE 5 (Continued)

	Log(Compliance Cost) (1)	Log γ (2)	Log ψ (3)
Regional water board fixed effects: ^a			
San Francisco Bay	-.408* (.216) [.969]	.033 (.061) [.271]	-.037 (.148) [.327]
Central Coast	-.539*** (.191) [1.240]	.040 (.036) [.270]	-.462*** (.122) [.247]
Los Angeles	-2.27*** (.191) [1.765]	.053 (.105) [.311]	.055 (.150) [.359]
Central Valley	-.015 (.252) [1.236]	.659*** (.058) [.261]	-.378*** (.075) [.227]
Colorado River	.276 (.317) [1.391]	.843*** (.098) [.305]	-.719*** (.160) [.358]
Santa Ana	.154 (.132) [1.751]	.347*** (.073) [.384]	-.125 (.140) [.470]
San Diego	-5.54*** (.848) [7.349]	-.050 (.105) [.431]	-.985*** (.302) [.674]
Constant	7.38*** (.434) [1.657]	7.88*** (.121) [.402]	.454*** (.160) [.350]
Adjusted R^2	.826	.829	.508

NOTE.—This table reports the ordinary least squares regression results of the logarithm of the estimated marginal compliance cost and the logarithm of the estimated regulator preferences for each of the 221 facilities active in the first quarter of 2005 on all facility attributes used in the estimation. Robust standard errors under the assumption that the estimated parameters are measured without error are given in parentheses, and the bootstrap standard errors without such an assumption are given in brackets. Asterisks are based on the former standard errors.

^a We omit region 1 (North Coast) in the regressions.

* $p < .10$.

** $p < .05$.

*** $p < .01$.

($\bar{a} = 1.08$), and (ii) the marginal benefit of compliance, as perceived by regulators for each facility, at the same negligence level, which we define as

$$\gamma_{\text{pre}}(\mathbf{x}_{i,t}) + \psi_{\text{pre}}(\mathbf{x}_{i,t})e'(\bar{a}|\mathbf{x}_{i,t}).$$

We find that there is a positive correlation between i and ii, but the variation in the expected penalties across facilities is not fully explained by the variation in the regulator preferences. This is partially because there are facilities with high (low) marginal compliance costs and low (high) marginal benefits of compliance, as can be seen in figure 3*B*. We show in corollary A2 in appendix B that it is optimal for the regulator to set a higher penalty for facilities with higher marginal compliance costs and/

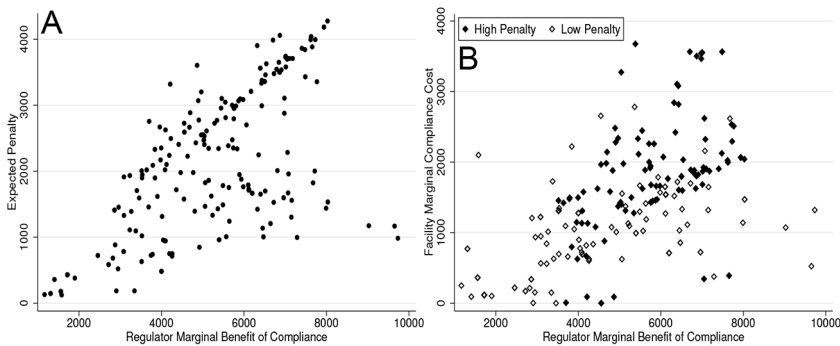


FIG. 3.—Disparities in penalties and regulator preferences. *A*, Scatterplot of the estimated marginal benefit of compliance for each facility at the average negligence level before the 2006 institutional changes ($\bar{a} = 1.08$) and the expected penalty at the same negligence level. *B*, Scatterplot of the estimated marginal benefit and cost of compliance for each facility. In *B*, we indicate whether the expected penalty at $\bar{a}$ is below or above the median, \$1,945.50.

or higher marginal benefits of compliance. Therefore, a facility with a low (high) benefit of compliance, as perceived by the regulator, may still face a stringent (lenient) penalty if its marginal compliance costs are high (low).

To quantitatively assess the extent to which the heterogeneity in regulator preferences explains the disparities in penalties associated with facility and local attributes, as documented in section III, we consider a counterfactual scenario where all facilities are subject to the regulator with the median values of γ and ψ (which we refer to as the median regulator). Row 1 of table 6 presents the counterfactual outcomes under the median regulator scenario compared with the baseline.⁴¹ Considering the expected penalties for each facility at the average negligence level before the 2006 institutional changes ($\bar{a} = 1.08$), we find that the standard deviation in the expected penalties across the facilities would, relative to the baseline scenario, decrease under the median regulator by a relatively small extent, 11%—and this decrease is not statistically significant.⁴² In addition, figure A3 provides evidence that this pattern is not limited to the penalties associated with a particular negligence level. These results suggest that the heterogeneity in the regulator preferences across facilities is not the main driver of the observed disparities in penalties, despite a

⁴¹ See table A8 for the corresponding results based on the estimated regulator preferences for the periods after the 2006 changes.

⁴² The 95% confidence interval of the change in the standard deviation of penalty stringency includes positive values. To see why this may happen in our model, consider a facility that faces a relatively low penalty but has high benefits of compliance. The median regulator would impose an even lower penalty for the same violation, intensifying the extent of the penalty dispersion.

TABLE 6
COUNTERFACTUAL ANALYSES: THE EFFECTS OF REGULATORY DISCRETION

	VIOLATION FREQUENCY		PENALTY	
	Mean	Standard Deviation	Mean	Disparity
(1) Median regulator	.055 [−.080, .174]	.278 [.029, .559]	−.119 [−.169, .010]	−.109 [−.518, .094]
(2) Uniform penalty	−.061 [−.130, −.018]	−.189 [−.304, −.097]	.013 [−.015, .115]	−1.000
(3) Linear penalty	−.059 [−.085, −.020]	−.050 [−.082, .015]	.158 [.100, .358]	.222 [−.032, .837]
(4) No enforcement costs	−.230 [−.330, −.169]	−.335 [−.379, −.144]	.485 [.224, .718]	.271 [−.168, 1.189]
(5) Green regulator	−.511 [−.779, −.494]	−.408 [−.693, −.371]	.766 [.172, 1.025]	−.807 [−.923, −.057]

NOTE.—This table presents the results of five counterfactual scenarios. For each scenario, we report the proportional changes, relative to the baseline scenario, in the following four statistics: the mean and the standard deviation of the equilibrium violation frequency across facilities, the mean of the equilibrium expected penalties, and the standard deviation of the equilibrium penalties at an identical negligence level ($\alpha = 1.08$) across facilities, as a measure of disparities in equilibrium penalty stringency. In row 1, every facility is under a regulator with preferences set at the median across all facilities active in the first quarter of 2005. In row 2, each facility is subject to the same penalty schedule. In row 3, regulators are constrained to impose a penalty schedule linear in the number of violations. In row 4, regulators do not bear any enforcement costs. The scenario in row 5 is similar to that of row 1, except that, from the perspective of the common regulator, damages from violation are high and enforcement costs are low (see the main text for details). The table reports the results based on the estimates for the period before the 2006 institutional changes. See also table A10 for the total expected costs borne by the regulator in each scenario.

large degree of heterogeneity in these preferences, as presented in tables 4 and 5.⁴³ Instead, variation in the distribution of compliance costs across the facilities seems to be responsible for the bulk of penalty disparities.

E. Limiting Regulatory Discretion

In our estimated model, we assume that the regulator knows the distribution of compliance costs faced by each facility. Moreover, we let her set the optimal enforcement schedule, as characterized in proposition 1, individually for each facility. That is, our model assumes that the regulator possesses expertise on the distribution of facilities’ compliance costs and is able to fully employ this expertise in the determination of the enforcement schedules. We now conduct counterfactual policy experiments in which we limit the regulator’s discretion to use her expertise. We evaluate the extent to which reducing regulatory discretion affects the equilibrium compliance

⁴³ To separately assess the role of heterogeneity in the external cost of violations (γ) vs. heterogeneity in enforcement costs (ψ), we consider two alternative hypothetical regulators in app. E.

behavior by the facilities, the penalties, and the enforcement costs borne by the regulator. With this goal, we consider two counterfactual scenarios. In the first one, the regulator must determine the same enforcement schedule to all facilities. This case is closely related to policies aimed at reducing the autonomy of the regional water boards, as discussed in section II.C. In the second case, the regulator can still vary the enforcement schedules across the facilities, but these schedules are constrained to be linear in the number of violations. In comparing the scenarios, our main variables of interest are the equilibrium negligence levels and penalties. In appendix E, we also provide, for each case, the total expected costs borne by the regulator, as defined in (4).

1. One-Size-Fits-All Policy

Suppose that the regulator sets a one-size-fits-all penalty schedule to minimize the sum of the total expected costs, as defined by (4), across all facilities. We refer to such a scenario as the uniform policy.⁴⁴ Relative to the penalties in the baseline scenario, the uniform policy would be harsher to some facilities and more lenient to others. Hence, in transitioning from the baseline to the uniform scenarios, some facilities would violate more and others less—so the aggregate impact of the uniform policy on violations would depend on the relative importance of these two effects. Row 2 of table 6 shows that, under the uniform policy, the average violation frequency across the facilities would fall by 6%. Despite this reduction, the total amount of penalties assigned would increase by 1%.

To evaluate in a more transparent manner how the constraint to a uniform policy affects enforcement costs, we consider penalty schedules that (i) are tailored to facility and local attributes and (ii) would achieve the same reduction of 6% in the average violation frequency across the facilities. In appendix D, we provide the details of one such schedule that would lower the total penalties by 2.5%, relative to the uniform policy.⁴⁵ This result illustrates that not being able to individually consider the attributes of each facility increases the total amount of penalties—and accordingly raises the enforcement costs—without improving compliance. From this perspective, our findings are consistent with the ones by Duflo et al. (2018), who find evidence favorable to providing regulators with discretion in targeting inspections.

⁴⁴ In considering a uniform policy, we restrict our attention to penalty schedules that are third-degree polynomials in the number of violations when there is any violation (and zero penalty for full compliance). See app. D for how we implement this counterfactual analysis.

⁴⁵ This particular set of penalty schedules is not very sophisticated, suggesting that a regulator with full discretion might be able to reduce total penalties even more, given a target violation rate. Row 3 of table A11 shows the other equilibrium objects under these schedules.

Another disadvantage of adopting the one-size-fits-all policy is that it limits the extent to which the preferences of local residents are represented in law enforcement. To see this, we regress whether a facility increases its negligence level in the transition from the baseline to the uniform policy on all control variables employed in the estimation of the structural model. Table 7 shows the regression results.⁴⁶ We find that switching to the uniform policy would lead to more violations by facilities that are major, pose a high threat to water quality, or are located in a county with a high average household income or a high approval rate for the 2006 proposition for water projects. These facilities tend to be perceived as having high benefits of compliance by the regulator (table 5) and, arguably, also by the local residents.

2. Linear vs. Nonlinear Penalties

According to our estimates, the enforcement schedule assigned to each facility is a strictly convex function of the negligence level (fig. 2). We now consider the scenario in which this schedule must be linear. Specifically, suppose that, for a facility with attributes vector $\mathbf{x}$, the regulator is constrained to set the penalty schedule $e(a|\mathbf{x}) = p(\mathbf{x})a$.⁴⁷ Row 3 of table 6 shows how the outcomes under linear penalties would compare with those in the baseline scenario. The convex penalties in the latter scenario allow the regulator to impose, to a given facility, large penalties per violation if the facility commits many violations (likely due to its draw of a high cost type, θ), while keeping the penalties relatively low in case the facility has a small number of violations (probably associated with a low θ). That is, holding $\mathbf{x}$ constant, convex penalties imply larger penalties per violation to facilities with relatively high compliance costs. If forced to assign linear penalties, the regulator would decrease (increase) the per-violation penalties for high (low) θ draws. Our findings indicate that the resulting decrease in violations by low θ facilities would more than compensate the increase by high θ ones, leading the expected violations to fall by roughly 6% overall. The equilibrium penalties would increase, on average, by 16%, raising enforcement costs.

The relatively mild reduction in violation frequencies under linear penalties contrasts with the large increases in penalty stringency—suggesting that, from the point of view of a regulator facing enforcement costs, linear penalties would be substantially less efficient than nonlinear ones. One way of assessing the magnitude of the excessive enforcement

⁴⁶ See table A9 for the counterpart results based on the post-2006 estimates.

⁴⁷ This linear penalty schedule (as a function of negligence level) can be implemented by $e(k|\mathbf{x}) = p(\mathbf{x})k$; with this, for a facility with negligence a and attributes $\mathbf{x}$, the expected penalty is $p(\mathbf{x})a$. See also app. D for details on the implementation of this counterfactual analysis.

TABLE 7
HETEROGENEOUS EFFECTS OF REGULATORY DISCRETION

	Dependent Variable: Increase in Negligence Level from Change to One-Size-Fits-All Policy?
Facility attributes:	
Major	.404*** (.061) [.098]
Started in 1982–87	–.118** (.054) [.055]
Started after 1987	–.214* (.109) [.079]
Advanced/tertiary	.153*** (.046) [.072]
Capacity utilization >87%	.148*** (.046) [.067]
Service population <10,000	–.007 (.063) [.093]
Special district	–.027 (.039) [.271]
High threat to water	.077* (.042) [.077]
Weather and pollution:	
Precipitation >75th percentile	.040 (.052) [.108]
Precipitation <25th percentile	–.092 (.116) [.0701]
Swimmable	.033 (.053) [.070]
County attributes:	
Irrigation use	–.042 (.066) [.244]
High income	.317*** (.059) [.141]
High density	–.082 (.053) [.097]
Low density	.390*** (.070) [.130]
High approval	.456*** (.071) [.172]

TABLE 7 (Continued)

	Dependent Variable: Increase in Negligence Level from Change to One-Size-Fits-All Policy?
Regional board fixed effects: ^a	
San Francisco	.556*** (.121) [.186]
Central Coast	.296*** (.110) [.229]
Los Angeles	-.514*** (.121) [.192]
Central Valley	.385*** (.085) [.138]
Colorado River	.642*** (.131) [.283]
Santa Ana	.611*** (.124) [.310]
San Diego	.244 (.180) [.399]
Constant	-.607*** (.169) [.229]
Adjusted R ²	.729

NOTE.—This table reports the ordinary least squares regression results where the dependent variable indicates whether the facility would increase its violation frequency under the uniform penalty scenario, compared with the baseline, before the 2006 institutional changes. The unit of observation is each of the 221 facilities active in the first quarter of 2005. Robust standard errors under the assumption that the estimated parameters are measured without error are given in parentheses, and the bootstrap standard errors without such an assumption are given in brackets. Asterisks are based on the former standard errors.

^a We omit region 1 (North Coast).

* $p < .10$.

** $p < .05$.

*** $p < .01$.

costs associated with linear penalties is through the following exercise: for every facility, we compute the linear penalty that would lead to the same expected number of violations as in the baseline scenario. Averaging across the facilities in the pre-2006 period, we find that penalties would increase by 10%.⁴⁸ These results stress the role played by nonlinear penalties in reducing the regulator’s enforcement costs. They are related

⁴⁸ See row 4 of table A11 for the other equilibrium outcomes under this scenario. Appendix D discusses how this analysis is conducted in detail.

to the findings by Blundell, Gowrisankaran, and Langer (2020), who, in the context of air quality regulation, argue that allowing the regulator to use relatively sophisticated dynamic penalty schemes leads to reductions in average fines.

F. Discussion

For each facility, we calculate the equilibrium negligence levels and penalties under two counterfactual scenarios in which, one at a time, we remove two key sources of inefficiencies in our model. We separately consider the cases with costless enforcement ($\psi = 0$) and no asymmetric information between the regulator and the facility. As discussed in section IV.C, either of these scenarios would lead to the first-best negligence levels, defined in (10). Since equilibrium penalties would be zero without asymmetric information, we report only the results of the scenario with no enforcement costs, in row 4 of table 6.

Mookherjee and Png (1994) show that optimal penalties for relatively mild offenses should, at the margin, be smaller than the corresponding social harm or external costs, while for sufficiently serious offenses, this relationship should revert. Row 4 of table 6 shows that, on average, the first effect prevails; the first-best average violation frequency is 23% lower than the baseline equilibrium. To induce such compliance levels, average penalties would increase by 49%.⁴⁹ In addition, figure A5 shows how the aforementioned gap between the marginal penalties and the marginal social harm widens as enforcement becomes more costly.⁵⁰

In the presence of costly enforcement and information frictions, should regulators have discretion? Insofar as regulator preferences reflect the preferences of local residents (sec. VI.C) and that the spillovers of water pollution to nearby areas are limited (to a distance of 20–25 miles; see Keiser and Shapiro 2019), policies that restrict regulatory discretion are suboptimal because the regulator cannot fully utilize her expertise on facilities' compliance costs to efficiently allocate enforcement resources. Furthermore, the extent to which law enforcement represents the preferences of local residents becomes limited. However, the regulator preferences might also reflect her private interests, possibly due to corruption or lack of dedication. These interests could lead the regulator to have inappropriately low perceptions of the benefits of compliance. Although our analysis cannot identify whether the heterogeneity in the estimated regulator preferences reflects differences in private or social concerns, we

⁴⁹ Table A10 shows that the compliance costs would be higher by 35%, the external costs would be lower by 25%, and the enforcement costs would be higher by 55% when the first-best negligence levels are induced, relative to the baseline scenario.

⁵⁰ In app. E, we consider a regulator whose marginal enforcement costs (ψ) are doubled for each facility. Row 5 of table A11 provides the equilibrium outcomes under this scenario.

compute a reasonable upper bound for the excess expected number of violations associated with private interests, if any.⁵¹

With this intent, we consider a counterfactual scenario where, as in the baseline case, a regulator sets a different penalty schedule to each facility. But instead of using our estimated regulator preferences, we consider a green regulator, who highly appreciates the benefits of compliance for all facilities. Specifically, the green regulator has $\gamma = \max\{\hat{\gamma}_{\text{pre}}(\mathbf{x}_{i,t})\} = 8,020$ and $\psi = 0.015$, which is the observed average staff costs per \$1 of penalty.⁵² The difference in violations frequencies between the baseline and the green regulator scenarios would be equal to the number of violations due to private interests, under the assumption that these interests are responsible for the entirety of the differences between the estimated regulator preferences and those of the green regulator. The comparison of these frequencies thus provides an upper bound for the share of violations associated with private concerns in the baseline scenario. Row 5 of table 6 shows that this upper bound is equivalent to 51% of the violations. These numbers are based on extreme assumptions about the green regulator preferences and should thus be interpreted cautiously.

VII. Conclusion

We provide an empirical framework to evaluate regulatory discretion by identifying and estimating a model of strategic interactions between a regulator and a privately informed discharger. Applying our framework to data on the regulation of wastewater treatment facilities in California, we estimate the environmental preferences and enforcement costs of regulators and the distribution of facilities' compliance costs. The estimates provide empirical support for regulatory discretion in our setting by showing that (i) regulator preferences reflect environmental preferences of local constituents, (ii) the heterogeneity of regulator preferences is not the main driver of observed disparities in penalties, and (iii) limiting regulatory discretion—by mandating a one-size-fits-all policy or a linear penalty schedule—would increase enforcement costs overall and increase violations by facilities with high marginal benefits of compliance as perceived by the regulator.

⁵¹ Alternatively, we could obtain the social costs of a violation by combining estimates of (i) the social cost of water pollution and (ii) the effects of a wastewater treatment facility's violation on water pollution in a nearby area. To measure i, we could employ existing estimates on the willingness to pay for water quality, but these estimates are likely to be exceeded by the social cost of water pollution, due to issues such as nonuse or existence values and the possibility that individuals may not be fully informed about the (health-related and otherwise) implications of local water quality changes (Keiser and Shapiro 2019). Estimating ii would require data on various sources of water pollution other than the violations by wastewater treatment facilities.

⁵² We observe staff costs for 28% of the penalty actions during the period of study in the data. The average ratio of staff costs to penalty amounts among these actions is 0.015.

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